

Social Security Design and Its Political Support

Online Appendix

Rasmus K. S. Berg

Martín Gonzalez-Eiras

Table of contents

About this appendix	3
Robustness at a glance	3
OA.1 The cross-section of figure 1.1	4
OA.1.1 Countries, values and sources	4
OA.1.2 Correlations	7
OA.2 Argentina	8
OA.2.1 Calibration	8
OA.2.2 The reform	9
OA.2.3 The reform across the intertemporal elasticity	10
OA.2.4 The equilibrium over pension designs	14
OA.2.5 The calibration across the intertemporal elasticity	16
OA.3 The rich economies with the design given	20
OA.3.1 Calibration	20
OA.3.2 The vector- X_i calibration	23
OA.3.3 The calibration across the intertemporal elasticity	24
OA.3.4 Counterfactuals on the U.S.	29
OA.3.5 The UK as host	38
OA.4 The endogenous pension design	46
OA.4.1 The cost and the cross-country test	46
OA.4.2 Counterfactuals on the U.S.	47
OA.4.3 The UK as host	53
OA.4.4 The design along the baseline path	57
OA.4.5 The timing of the design choice	61
OA.4.6 An alternative cost specification	61
OA.5 Numerical checks	63
OA.5.1 Date-specific and stationary policies	63
OA.5.2 Equilibrium selection	64
OA.6 Replication	66
References	68

About this appendix

This online appendix accompanies *Social Security Design and Its Political Support* by Rasmus K. S. Berg and Martín Gonzalez-Eiras. It holds the tables and figures behind the paper's quantitative sections, the robustness checks that the paper's appendix G summarizes in one figure, and the numerical checks behind its results, with each figure set beside the tables that print its numbers. Every exhibit is built by the paper's own pipeline from the model's results, so the numbers are the paper's to the last printed digit; [replication](#) lists where each one comes from.

Reading this edition. This is the print edition of the online appendix. It holds every table and figure that is not in the paper, numbered Table or Figure OA.*k*; where the site shows one of the paper's own exhibits beside them, a line names it and the section of the paper that carries it. Where an exhibit exists under both calibrations of the households' taste for leisure, the paper's calibration, common X , comes first and the alternative, vector X_i (OA.3.2), follows it. The site adds, for every exhibit, the value under the pointer, the link from a marker of a figure to the table row that prints it, the switch between the two calibrations with the cells they differ in marked, and the download of the files each exhibit reads.

Robustness at a glance

Figure G.1 of the paper, repeated here, sets its quantitative results against every specification the paper reports. Each panel is one result: how the tax rate responds to the Bismarckian index θ , to acute ageing and to France's income distribution and voting patterns with the design given, and how the chosen design responds to the last three. Each marker is the same result under one specification: the intertemporal elasticity of substitution (IES) ρ , the calibration of the taste for leisure, the host economy, and for the chosen design the cost specification and the Frisch elasticity ξ . Results are changes against the host's own baseline at the same IES, and the paper's own reading, the U.S. under a common X at an IES of one with the cost of section 7, is the highlighted marker.

The site shows here the paper's figure G.1 (appendix G), which this edition does not reprint.

Figure G.1 of the paper: a marker across the zero line from the highlighted one reverses the paper's sign. With the design given, the Beveridgean corner against the Bismarckian one, acute ageing and France's voting patterns raise the tax rate in every specification; France's income distribution lowers it on the U.S., and on the UK raises it under a common X and lowers it under the vector X_i . With the design chosen, acute ageing makes the design more Bismarckian and France's voting patterns less so throughout, while France's income distribution moves it in a direction that changes with the IES and the host, and to the Bismarckian corner under the cost on the design.

Built on 2026-10-07 from commit [b88d83f](#).

OA.1 The cross-section of figure 1.1

Figure 1.1 of the paper sets pension spending and the Bismarckian–Beveridgean index of the OECD countries against their population growth and their income inequality. This chapter gives the values behind each marker, the year each refers to, the sources, and the correlations the paper’s introduction describes.

OA.1.1 Countries, values and sources

The table of countries prints every variable the figure draws, with the year of each value; the table of sources names, for every column, the provider, the dataset and the series, and when it was retrieved.

Beside these, the site shows the paper’s figure 1.1 (section 1), which this edition does not reprint.

Figure 1.1 of the paper: one marker per country, with the three economies the paper calibrates labeled. The Bismarckian–Beveridgean index is the ratio of the gross replacement rates for men at mean and at half-mean earnings: one where benefits are proportional to earnings, one half where they are flat. Pension spending and the index both fall as population growth rises, and neither has a clear relation to the Gini coefficient (table OA.3).

Table OA.1: The cross-section of figure 1.1 of the paper, by country

Country	Pension spend- ing, % of GDP	Year	Population 2020 / 1990	Bismarckian Beveridgean index	Gini, dispos- able income	Year
Australia	4.2	2020	1.503	0.499	0.340	2020
Austria	14.3	2020	1.161	1.000	0.304	2020
Belgium	11.5	2020	1.155	0.643	0.274	2020
Canada	6.4	2020	1.373	0.729	0.318	2020
Czech Republic	8.6	2020	1.033	0.603	0.254	2020
Denmark	8.2	2020	1.133	0.639	0.283	2020
Finland	12.6	2020	1.109	1.000	0.277	2020
France	14.4	2020	1.192	1.000	0.321	2020
Germany	11.2	2020	1.047	0.892	0.323	2020
Greece	17.9	2020	1.049	0.857	0.331	2020
Hungary	7.7	2020	0.932	1.000	0.303	2020
Iceland	3.0	2020	1.398	0.711	0.275	2020
Ireland	3.3	2020	1.435	0.500	0.302	2020
Italy	17.4	2020	1.048	1.000	0.358	2020
Japan	9.6	2020	1.020	0.750	0.330	2020
Korea	3.5	2020	1.209	0.724	0.333	2020
Luxembourg	9.0	2020	1.651	0.847	0.341	2020
Mexico	4.6	2020	1.517	0.756	0.456	2020
Netherlands	6.8	2020	1.167	0.954	0.295	2020
New Zealand	5.1	2020	1.496	0.604	0.320	2020
Norway	7.6	2020	1.268	0.759	0.277	2020
Poland	11.4	2020	1.004	0.962	0.308	2020
Portugal	13.6	2020	1.040	0.982	0.328	2020
Spain	12.8	2020	1.218	1.000	0.343	2020
Sweden	8.5	2020	1.210	0.868	0.307	2020
Switzerland	6.9	2020	1.287	0.831	0.343	2020
Turkey	7.4	2020	1.513	1.000	0.434	2020
United Kingdom	7.8	2020	1.166	0.694	0.329	2020
United States	7.4	2020	1.328	0.790	0.419	2020

Note: The 29 OECD members plotted in figure 1.1 of the paper. Each variable is read in 2020 or the nearest year within 2 years of it (the year columns); population growth is the population of 2020 over that of 1990. Sources in table OA.2. In bold the economies the paper calibrates.

Table OA.2: The sources of figure 1.1 of the paper

Columns	Source	Series and years	Note
<i>Plotted in the figure of the paper</i>			
pensionSpending	OECD: Social Expenditure Database, OECD.ELS.SPD:DSD_SOCASHAG@DF_SOPX_AG (paid), Data Explorer; retrieved 2026-09-30	public (ES10), old age and survivors (TP01), in XAG@DF_SOPX_AG (paid), Years: 2020 (29 of 29)	The OECD's 'pension spending': cash expenditure on old-age and survivors pensions (Pensions at a Glance measure PEP to its rounding).
pop1990_oecd, pop2020_oecd, popGrowth	OECD: Historical population data, OECD.ELS.SAE:DSD_POPHISTON@DF_POP_HIST (paid), Data Explorer; retrieved 2026-09-30	POP, persons, both sexes, all ages, historical; pop2020/pop1990. Years: 1990 and 2020 (29 of 29)	France: the series adds the overseas territories from 1991, so its ratio (1.192) is not on a constant territory (World Bank 1.160).
bbIndex	derived: Pensions at a Glance 2021, table 4.1	grr100_men / grr050_men. Years: 2020 (29 of 29)	Men: the table headline and the concept of section 6.
gini, gini_year, gini_wiidId, gini_source	UNU-WIDER: World Income Inequality Database (WIID), version 8 September 2026, doi 10.35188/UNU-WIDER/WIID-080926; version 8 Sep 2026; retrieved 2026-09-30	the observation WIID flags for its Companion (wiidcompanion = Yes): disposable income, persons, per capita except where gini_source says otherwise; 0-1 scale. Years: 2020 (29 of 29)	gini_wiidId is the WIID row id.
<i>Further concepts in data/oecdFigure1.csv</i>			
pensionSpending_cashInKind	OECD: Social Expenditure Database, OECD.ELS.SPD:DSD_SOCASHAG@DF_SOPX_AG (paid), Data Explorer; retrieved 2026-09-30	as pensionSpending, cash plus in kind (_T). Years: 2020 (29 of 29)	The in-kind part is services to the elderly (care, home help), largest in NOR 2.57, SWE 2.43, DNK 1.87, FIN 1.61.
pop1990_wb, pop2020_wb, popGrowth_wb	World Bank: World Development Indicators; API lastupdated 2026-07-13; retrieved 2026-09-30	SP.POP.TOTL; popGrowth_wb = pop2020/pop1990. Years: 1990 and 2020 (29 of 29)	—
grr050_men, grr100_men, grr050_women, grr100_women	OECD: Pensions at a Glance 2021, table 4.1; StatLink xlsx “Version 1 - Last updated: 08-Dec-2021”; retrieved 2026-09-30	gross pension replacement rate at 0.5 and 1.0 times average earnings, mandatory schemes, % of individual earnings, one decimal as published; women = men where the table gives none. Years: pension rules in 2020 (29 of 29)	The OECD Data Explorer dataflows DSD_PAG@DF_PRR and DSD_PAG@DF_PAG now hold only 2024 (Pensions at a Glance 2025); the publication StatLink still serves the 2020 vintage.
bbIndex_women	derived: Pensions at a Glance 2021, table 4.1	grr100_women / grr050_women. Years: 2020 (29 of 29)	—
bbIndex_mean	derived: Pensions at a Glance 2021, table 4.1	(bbIndex + bbIndex_women) / 2. Years: 2020 (29 of 29)	—
gini_wiidMarket, gini_wiidMarket_year	UNU-WIDER: WIID, version 8 September 2026; version 8 Sep 2026; retrieved 2026-09-30	source OECD.Stat (Income Distribution Database), market income, square-root scale, persons; 0-1 scale. Years: 2020 (22 of 29); 2018: JPN; 2019: DNK, FRA, DEU, CHE, TUR;	Before taxes and transfers; retirees without market income count at zero.

OA.1.2 Correlations

The table prints the pairwise correlations between the four variables of figure 1.1, Pearson and Spearman, with the number of countries that report both.

Table OA.3: Correlations in the cross-section of figure 1.1 of the paper

Variables	n	Pearson r	p	Spearman	p
Pension spending, % of GDP and Population 2020 / 1990	29	−0.58	0.001	−0.62	< 0.001
Pension spending, % of GDP and Bismarckian–Beveridgean index	29	0.63	< 0.001	0.60	0.001
Pension spending, % of GDP and Gini, disposable income	29	−0.08	0.681	0.00	0.992
Population 2020 / 1990 and Bismarckian–Beveridgean index	29	−0.39	0.039	−0.30	0.112
Population 2020 / 1990 and Gini, disposable income	29	0.43	0.020	0.34	0.068
Bismarckian–Beveridgean index and Gini, disposable income	29	0.18	0.346	0.23	0.231

Note: The concepts plotted in figure 1.1 of the paper. Pairwise complete observations; the p -values are those of the two-sided tests of zero correlation. Variables as in table OA.1 and their sources in table OA.2.

OA.2 Argentina

Section 5 of the paper calibrates the model with informal households to Argentina before the coverage reforms and reads the reforms as a rise of the universal pension ϵ ; appendix E gives the survey, the identifying formulas and the sources. This chapter holds the calibration under both treatments of the taste for leisure, the reform at every intertemporal elasticity of the model's grid, the equilibrium over pension designs, and the calibration at every IES.

OA.2.1 Calibration

The paper takes the taste for leisure common to the four formal quartiles and pins its level by the average formal workweek, so relative hours are a prediction. The alternative identifies one X_i per quartile from the survey's relative hours. Only the productivity and leisure parameters differ between the two: the aggregate targets, and every result of section 5, are the same under both.

Beside these, the site shows the paper's table 1 (section 5), which this edition does not reprint.

Table OA.4: Calibration, Argentina (vector X_i calibration)

Parameter	Value	Target
ϵ (<i>pre-reform</i>)	0.21	Basic pension, 70% of the minimum, discounted
θ	0.84	Replacement rate dispersion
α	0.35	Factor income shares
ν_t	[1.65, 0.96]	30-year gross population growth rates
ξ	0.30	Elasticity of labor supply
β	0.65	Capital-output ratio of 3.23
X_i	[0.74, 0.77, 0.88, 1.24]	Relative working hours
η_i	[0.51, 0.72, 0.95, 1.69]	Income distribution
γ_0	0.47	Elderly without a pension, 32% of all households
ω	1.45	Social security tax of 10.9%
η_0	0.288	Informal relative income, equation (E.1) of the paper
X_0	0.375	Informal relative working hours

Note: Our default specification relies on $\rho = 1$. Vector- X_i calibration: X_i is identified from relative hours, which are data here, and the level of $\bar{h}$ is then not identified — only its ratio to the baseline is. β , ω , the tax rate and the savings rate are common to the two variants; what differs is anything defined through η or X .

OA.2.2 The reform

The first table is the paper's: the effect in 2010 of raising ϵ to the universal level, with the tax rate held at its pre-reform path and chosen politically. The second prints the paths of the tax rate, the savings rate and the workweek before and after the reform.

Beside these, the site shows the paper's table 2 (section 5), which this edition does not reprint.

Table 2 of the paper: the effect in 2010 of raising the universal pension ϵ to the level at which informal retirees receive the benefit of the least productive formal workers, unanticipated and permanent. The first row is the pre-reform economy; the second holds the tax rate at its pre-reform path and so isolates the response of households; the third lets the electorate reset the tax. Taxes rise, because the reform moves benefits toward informal retirees, who are poorer than formal ones and value them more at the margin, and savings and hours fall a little; with the tax held fixed, formal savings rise instead, since informal households now hold a claim on the system and save less.

Under the vector- X_i calibration this exhibit is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so the aggregates the reform moves do not depend on how the taste for leisure is split across the quartiles.

Table OA.5: Pension system reform, year 2010 (vector X_i calibration)

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	10.92%	14.40%	42.54
Economic Equilibrium	10.92%	14.73%	42.46
Full effect	13.00%	13.97%	42.29

Note: The reform permanently shifts $\epsilon = 1 - \theta + \theta\eta_1 h_1/h$ at the calibration year, unanticipated. The “Economic Equilibrium” scenario solves for the new economic equilibrium path with taxes as in the Baseline scenario, but with the new permanent ϵ ; the “Full effect” scenario lets taxes be determined by the politico-economic equilibrium. $\rho = 1$. The savings rate is savings relative to GDP. Vector- X_i calibration: see the note to table OA.4.

The same reform along the path: the tax rate, the savings rate and the workweek in every model period from the reform year on, on the pre-reform path and on the reform path side by side. The tax response builds over the first periods, because informal households, now covered, save less for retirement and as retirees demand more taxation; section 5 of the paper quotes the response one period on.

Under the vector- X_i calibration this exhibit is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so the aggregates the reform moves do not depend on how the taste for leisure is split across the quartiles.

Table OA.6: Pension system reform along the path, Argentina

Year	Tax rate		Savings rate		Avg. workweek (hours)	
	Pre-reform	Reform	Pre-reform	Reform	Pre-reform	Reform
2010	10.92%	13.00%	14.40%	13.97%	42.54	42.29
2040	18.88%	22.41%	12.50%	12.03%	42.19	41.53
2070	24.01%	26.78%	11.78%	11.44%	41.78	41.10
2100	23.36%	25.84%	11.87%	11.56%	41.73	41.10
2130	23.46%	26.04%	11.84%	11.49%	41.70	41.08
2160	23.63%	26.55%	11.19%	10.65%	41.82	41.17

Note: The reform of table 2 of the paper, with taxes determined by the politico-economic equilibrium, at $\rho = 1$. The pre-reform columns are the equilibrium path without the reform. The savings rate is savings relative to GDP. The workweek is normalised so that the pre-reform path reads the observed average workweek in 2010. The terminal period of the finite horizon, where savings are zero, is not shown.

Table OA.7: Pension system reform along the path, Argentina (vector X_i calibration)

Year	Tax rate		Savings rate		Avg. workweek (hours)	
	Pre-reform	Reform	Pre-reform	Reform	Pre-reform	Reform
2010	10.92%	13.00%	14.40%	13.97%	42.54	42.29
2040	18.88%	22.41%	12.50%	12.03%	42.19	41.53
2070	24.01%	26.78%	11.78%	11.44%	41.78	41.10
2100	23.36%	25.84%	11.87%	11.56%	41.73	41.10
2130	23.46%	26.04%	11.84%	11.49%	41.70	41.08
2160	23.63%	26.55%	11.19%	10.65%	41.82	41.17

Note: The reform of table OA.5, with taxes determined by the politico-economic equilibrium, at $\rho = 1$. The pre-reform columns are the equilibrium path without the reform. The savings rate is savings relative to GDP. The workweek is normalised so that the pre-reform path reads the observed average workweek in 2010. The terminal period of the finite horizon, where savings are zero, is not shown. Vector- X_i calibration: see the note to table OA.4.

OA.2.3 The reform across the intertemporal elasticity

Section 5 reads the reform across the intertemporal elasticity of substitution (IES) ρ from the figure, which is the paper's; the model is solved on a grid of sixteen values. The table prints the reform at every point of the grid.

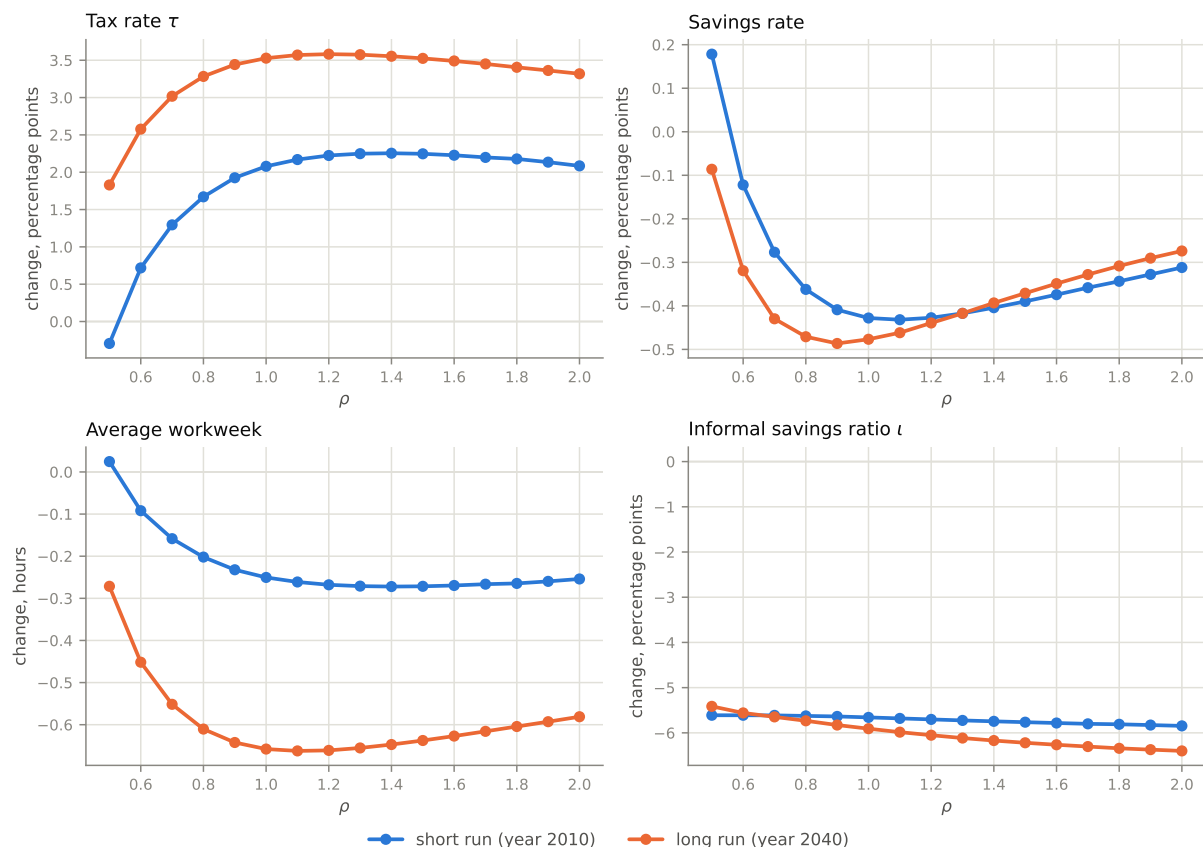
Beside these, the site shows the paper's figure 5.2 (section 5), which this edition does not reprint.

Figure 5.2 of the paper: the effect of the reform on the tax rate, the savings rate, the workweek and the ratio of informal to formal savings, on impact and one period on, at every IES of the grid, each economy recalibrated to the same pre-reform targets. Below an IES of one the responses are dampened, and at the lowest elasticities the tax response on impact reverses, since the young

resist taxation more and the calibration needs a heavier political weight of the old to reproduce the observed tax rate; between one and two the responses stay close to the log benchmark.

Under the vector- X_i calibration this exhibit is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so the aggregates the reform moves do not depend on how the taste for leisure is split across the quartiles.

Figure OA.1: Short- and long-run effects of the pension reform as a function of ρ , vector- X_i calibration



Note: The paper's figure under the vector- X_i calibration (section OA.3.2); its baseline levels are the baseline rows of the tables of this section.

The values behind the figure's short-run series, one row per point of the grid: the post-reform tax rate, the change in the savings rate and the workweek in the reform year. The pre-reform row is common to every IES, because each is recalibrated to the same targets; the pre-reform savings rate is not a target and varies with the IES, so its column is a change.

Under the vector- X_i calibration this exhibit is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so the aggregates the reform moves do not depend on how the taste for leisure is split across the quartiles.

Table OA.8: Pension system reform, year 2010, function of ρ

Scenario	Tax rate	Change in savings rate	Avg. workweek (hours)
Pre-reform	10.92%	–	42.54
Post-reform, $\rho = 0.5$	10.63%	+0.18 p.p.	42.56
Post-reform, $\rho = 0.6$	11.64%	–0.12 p.p.	42.45
Post-reform, $\rho = 0.7$	12.22%	–0.28 p.p.	42.38
Post-reform, $\rho = 0.8$	12.59%	–0.36 p.p.	42.34
Post-reform, $\rho = 0.9$	12.85%	–0.41 p.p.	42.31
Post-reform, $\rho = 1.0$	13.00%	–0.43 p.p.	42.29
Post-reform, $\rho = 1.1$	13.09%	–0.43 p.p.	42.28
Post-reform, $\rho = 1.2$	13.15%	–0.43 p.p.	42.27
Post-reform, $\rho = 1.3$	13.17%	–0.42 p.p.	42.27
Post-reform, $\rho = 1.4$	13.18%	–0.40 p.p.	42.27
Post-reform, $\rho = 1.5$	13.17%	–0.39 p.p.	42.27
Post-reform, $\rho = 1.6$	13.15%	–0.37 p.p.	42.27
Post-reform, $\rho = 1.7$	13.12%	–0.36 p.p.	42.27
Post-reform, $\rho = 1.8$	13.10%	–0.34 p.p.	42.27
Post-reform, $\rho = 1.9$	13.06%	–0.33 p.p.	42.28
Post-reform, $\rho = 2.0$	13.01%	–0.31 p.p.	42.28

Note: The reform permanently shifts $\epsilon = 1 - \theta + \theta\eta_1 h_1/h$. Each ρ is separately recalibrated to the same pre-reform targets, so the pre-reform tax rate and workweek are common to every ρ . The savings rate is savings relative to GDP; its pre-reform level is not targeted (the capital–output ratio is) and varies with ρ , so each post-reform row reports the change against its own ρ 's pre-reform level in percentage points.

Table OA.9: Pension system reform, year 2010, function of ρ (vector X_i calibration)

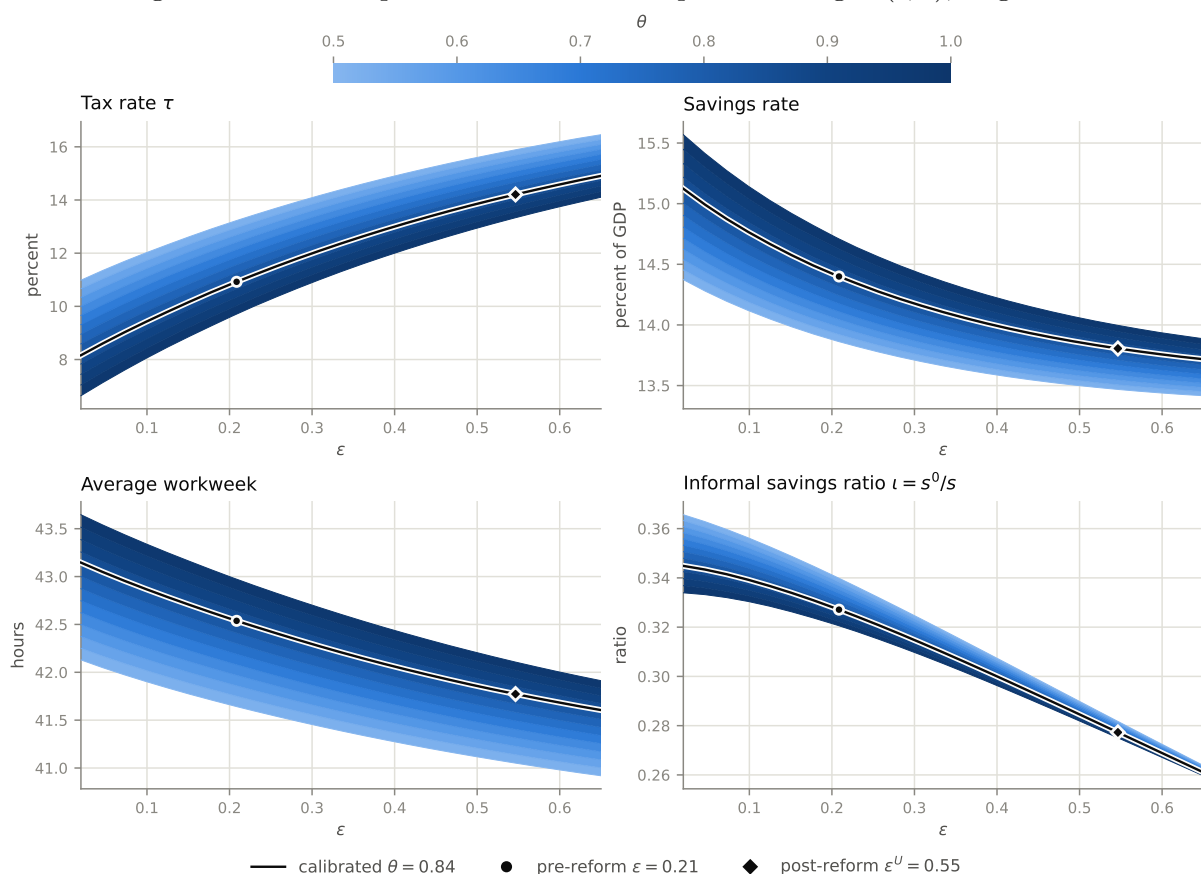
Scenario	Tax rate	Change in savings rate	Avg. workweek (hours)
Pre-reform	10.92%	–	42.54
Post-reform, $\rho = 0.5$	10.63%	+0.18 p.p.	42.56
Post-reform, $\rho = 0.6$	11.64%	–0.12 p.p.	42.45
Post-reform, $\rho = 0.7$	12.22%	–0.28 p.p.	42.38
Post-reform, $\rho = 0.8$	12.59%	–0.36 p.p.	42.34
Post-reform, $\rho = 0.9$	12.85%	–0.41 p.p.	42.31
Post-reform, $\rho = 1.0$	13.00%	–0.43 p.p.	42.29
Post-reform, $\rho = 1.1$	13.09%	–0.43 p.p.	42.28
Post-reform, $\rho = 1.2$	13.15%	–0.43 p.p.	42.27
Post-reform, $\rho = 1.3$	13.17%	–0.42 p.p.	42.27
Post-reform, $\rho = 1.4$	13.18%	–0.40 p.p.	42.27
Post-reform, $\rho = 1.5$	13.17%	–0.39 p.p.	42.27
Post-reform, $\rho = 1.6$	13.15%	–0.37 p.p.	42.27
Post-reform, $\rho = 1.7$	13.12%	–0.36 p.p.	42.27
Post-reform, $\rho = 1.8$	13.10%	–0.34 p.p.	42.27
Post-reform, $\rho = 1.9$	13.06%	–0.33 p.p.	42.28
Post-reform, $\rho = 2.0$	13.01%	–0.31 p.p.	42.28

Note: The reform permanently shifts $\epsilon = 1 - \theta + \theta\eta_1 h_1/h$. Each ρ is separately recalibrated to the same pre-reform targets, so the pre-reform tax rate and workweek are common to every ρ . The savings rate is savings relative to GDP; its pre-reform level is not targeted (the capital–output ratio is) and varies with ρ , so each post-reform row reports the change against its own ρ 's pre-reform level in percentage points. Vector- X_i calibration: see the note to table OA.4.

OA.2.4 The equilibrium over pension designs

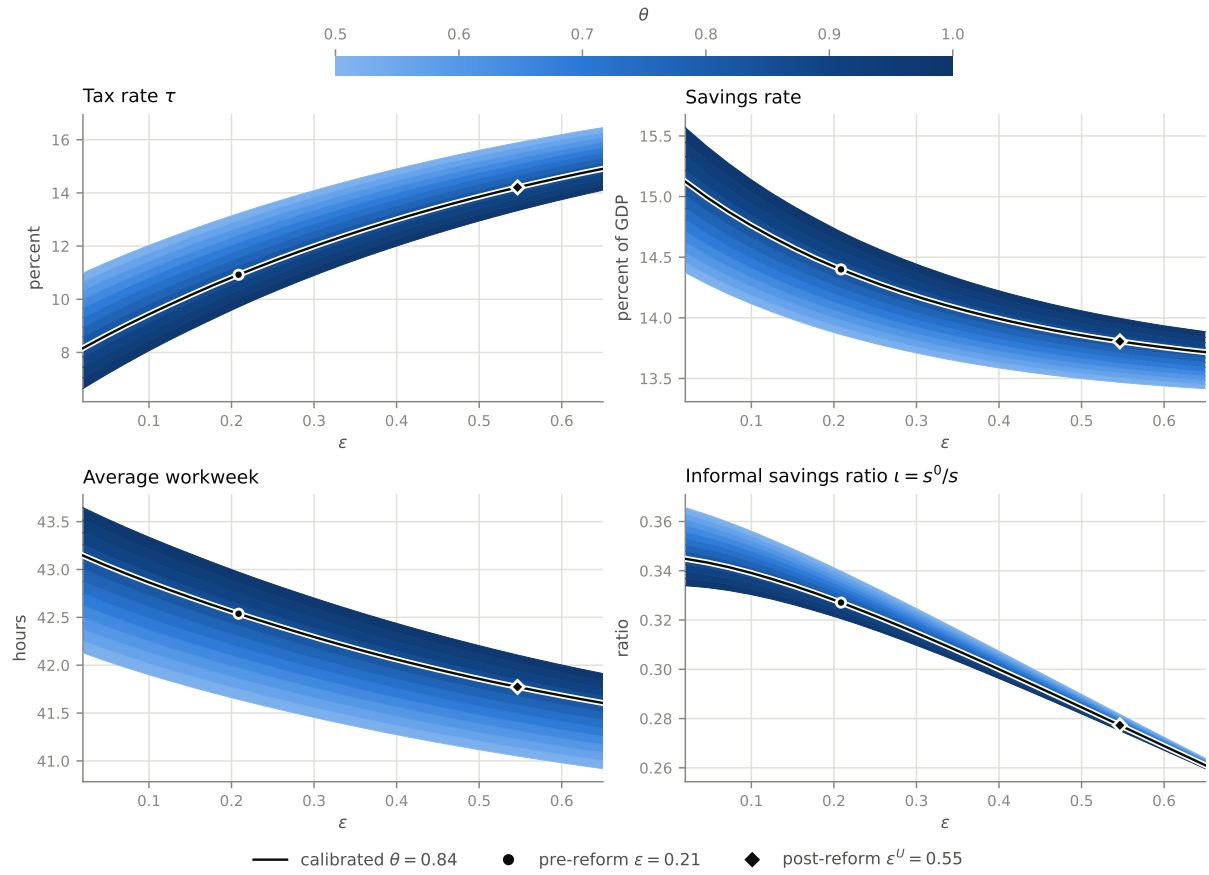
The equilibrium in 2010 over a grid of designs (ϵ, θ) , each point an economy that has always had that design, with the pre-reform calibration and the universal system at the calibrated θ marked. The tax rate rises with ϵ and falls with θ , the savings rate and the workweek fall with ϵ and rise with θ , and the savings of informal relative to formal households fall with ϵ . Between the two markers the tax rate rises by more than it does on impact in section 5 of the paper, because the savings of informal households have adjusted to the universal system; the difference is close to the reform's effect by 2040 (OA.2.2).

Figure OA.2: The equilibrium in 2010 over pension designs (ϵ, θ) , Argentina



Note: The line is the calibrated θ , the markers the pre-reform ϵ and the universal level of the reform. The savings rate is savings relative to GDP.

Figure OA.3: The equilibrium in 2010 over pension designs (ϵ, θ) , Argentina, vector- X_i calibration



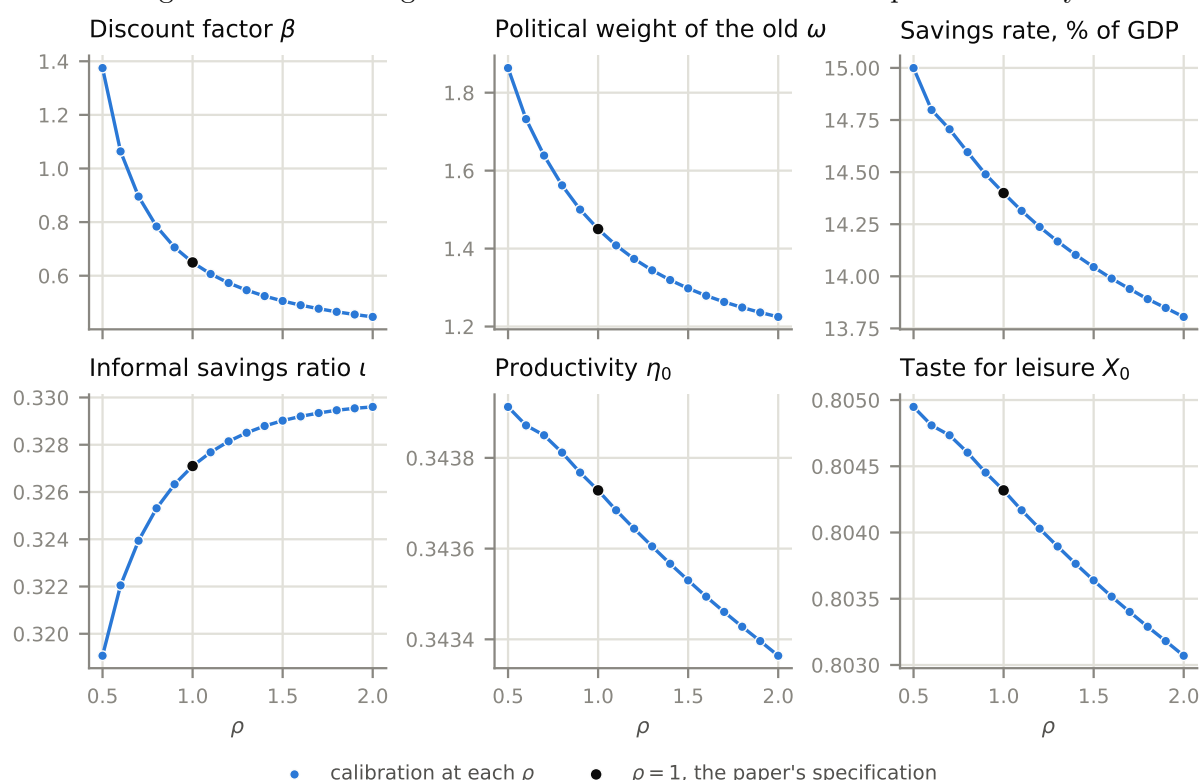
Note: The line is the calibrated θ , the markers the pre-reform ϵ and the universal level of the reform. The savings rate is savings relative to GDP.

OA.2.5 The calibration across the intertemporal elasticity

The table and the figure print the parameters solved jointly with the equilibrium path at every IES of the grid, which show how the calibration absorbs a change in ρ : matching the targets at an IES of 0.6 or below takes a thirty-year discount factor above one.

Four panels are parameters solved at each IES and two are predictions of the calibration, the savings rate and the informal savings ratio in 2010. The discount factor and the political weight of the old both fall as the IES rises, the second because the young resist taxation less and a lighter weight of the old reproduces the observed tax rate; the savings rate falls, the informal savings ratio rises, and the informal productivity and taste for leisure barely move.

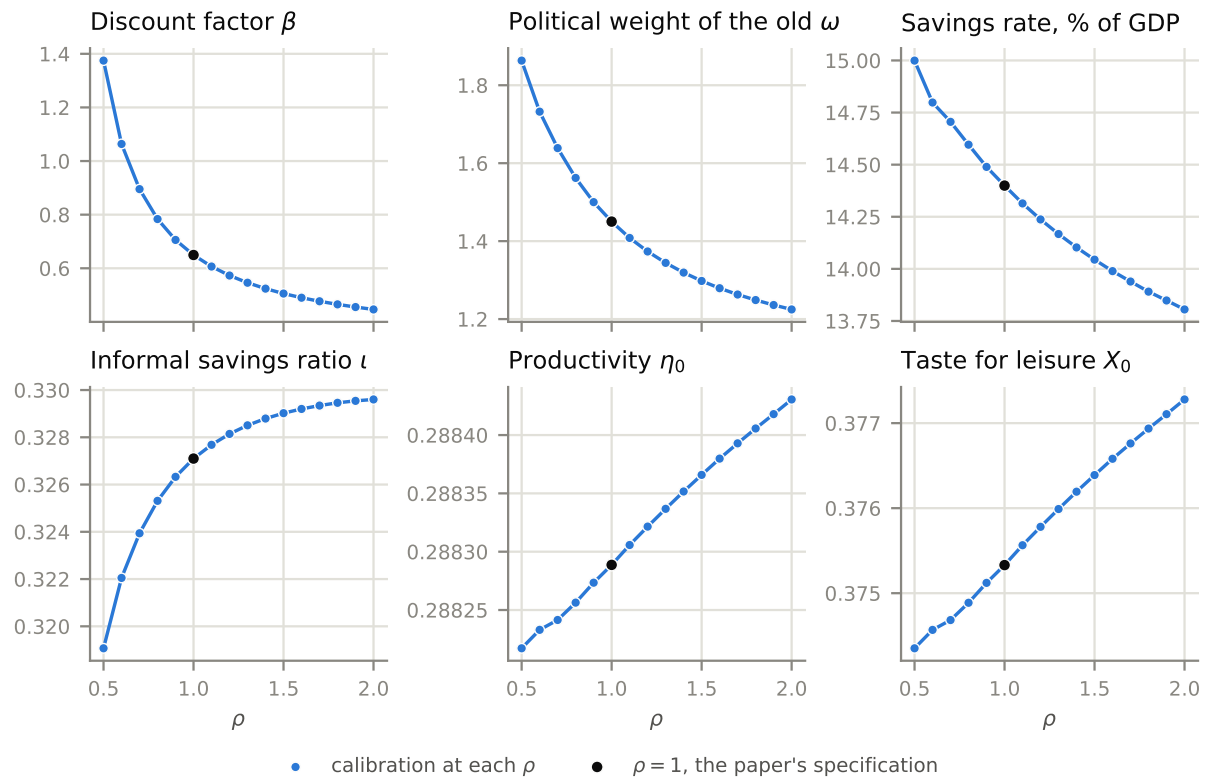
Figure OA.4: The Argentine calibration across the intertemporal elasticity



Note: The parameters solved jointly with the equilibrium path at each IES of the grid, against the same targets; the table of this section prints their values, and $\rho = 1$ is the paper's calibration.

Under the vector- X_i calibration the discount factor, the political weight of the old, the savings rate and the informal savings ratio are those of the common X at every IES; only the informal productivity and taste for leisure take other values, and they too barely move with the IES.

Figure OA.5: The Argentine calibration across the intertemporal elasticity, vector- X_i calibration



Note: The parameters solved jointly with the equilibrium path at each IES of the grid, against the same targets; the table of this section prints their values, and $\rho = 1$ is the paper's calibration.

Table OA.10: The calibration across ρ , Argentina

CRRA (ρ)	β	ω	η_0	X_0	X	Savings rate	ι
0.5	1.375	1.863	0.344	0.805	1.94	15.00%	0.319
0.6	1.064	1.732	0.344	0.805	1.94	14.80%	0.322
0.7	0.895	1.639	0.344	0.805	1.94	14.71%	0.324
0.8	0.784	1.562	0.344	0.805	1.94	14.60%	0.325
0.9	0.706	1.500	0.344	0.804	1.94	14.49%	0.326
1.0	0.650	1.450	0.344	0.804	1.94	14.40%	0.327
1.1	0.607	1.408	0.344	0.804	1.93	14.31%	0.328
1.2	0.573	1.373	0.344	0.804	1.93	14.24%	0.328
1.3	0.546	1.344	0.344	0.804	1.93	14.17%	0.329
1.4	0.524	1.319	0.344	0.804	1.93	14.10%	0.329
1.5	0.506	1.298	0.344	0.804	1.93	14.04%	0.329
1.6	0.490	1.279	0.343	0.804	1.93	13.99%	0.329
1.7	0.477	1.263	0.343	0.803	1.93	13.94%	0.329
1.8	0.465	1.249	0.343	0.803	1.93	13.89%	0.329
1.9	0.455	1.236	0.343	0.803	1.92	13.85%	0.330
2.0	0.446	1.225	0.343	0.803	1.92	13.81%	0.330

Note: Each ρ is calibrated to the same targets, a capital–output ratio of 3.23 and a tax rate of 10.92% in 2010; the calibration at $\rho = 1$ is table 1 of the paper. The savings rate (savings relative to GDP) and the informal savings ratio $\iota = s^0/s$ are those of 2010, predictions rather than targets.

Table OA.11: The calibration across ρ , Argentina (vector X_i calibration)

CRRA (ρ)	β	ω	η_0	X_0	X	Savings rate	ι
0.5	1.375	1.863	0.288	0.374	—	15.00%	0.319
0.6	1.064	1.732	0.288	0.375	—	14.80%	0.322
0.7	0.895	1.639	0.288	0.375	—	14.71%	0.324
0.8	0.784	1.562	0.288	0.375	—	14.60%	0.325
0.9	0.706	1.500	0.288	0.375	—	14.49%	0.326
1.0	0.650	1.450	0.288	0.375	—	14.40%	0.327
1.1	0.607	1.408	0.288	0.376	—	14.31%	0.328
1.2	0.573	1.373	0.288	0.376	—	14.24%	0.328
1.3	0.546	1.344	0.288	0.376	—	14.17%	0.329
1.4	0.524	1.319	0.288	0.376	—	14.10%	0.329
1.5	0.506	1.298	0.288	0.376	—	14.04%	0.329
1.6	0.490	1.279	0.288	0.377	—	13.99%	0.329
1.7	0.477	1.263	0.288	0.377	—	13.94%	0.329
1.8	0.465	1.249	0.288	0.377	—	13.89%	0.329
1.9	0.455	1.236	0.288	0.377	—	13.85%	0.330
2.0	0.446	1.225	0.288	0.377	—	13.81%	0.330

Note: Each ρ is calibrated to the same targets, a capital–output ratio of 3.23 and a tax rate of 10.92% in 2010; the calibration at $\rho = 1$ is table OA.4. The savings rate (savings relative to GDP) and the informal savings ratio $\iota = s^0/s$ are those of 2010, predictions rather than targets. The vector- X_i calibration has no common X . Vector- X_i calibration: see the note to table OA.4.

OA.3 The rich economies with the design given

Section 6 of the paper calibrates the model without informal households to the U.S., France and the UK, and changes the U.S. calibration one characteristic at a time, its pension design, its demography and the characteristics that set France apart; appendix F gives the data and the household heterogeneity behind the calibration. This chapter holds every table behind section 6's figure, the counterfactuals under CRRA preferences and the same counterfactuals with the UK as host, each under both calibrations of the taste for leisure, and the calibrations at every intertemporal elasticity of substitution (IES) of the model's grid.

OA.3.1 Calibration

The first table prints the country parameters, with those the three countries share in a lower panel; the others print, by income group, the population shares, the taste for leisure, the productivities and the propensities to vote. France is grouped at the U.S. income percentiles and, for the UK as host, at the UK's. The UK is also regrouped at the U.S. percentiles for one row of the paper's cross-country test in section 7; that regrouping is a linear fit of the UK's own group means against their cumulative population shares, not a recount of the survey.

Beside these, the site shows the paper's table 3 (section 6); table 6 (appendix F); table 7 (appendix F); table 8 (appendix F); table 9 (appendix F), which this edition does not reprint.

Table 3 of the paper: the three calibrations at an IES of one. France has the lowest population growth and the narrowest productivity spread of the three and the U.S. the highest population growth, while the political weight of the old is close in the U.S. and France and lower in the UK.

Table 6 of the paper: the U.S. households in three income groups, the first two cut so that their mean incomes are half the mean and the mean. Productivity and the propensity to vote both rise with income, and the low group holds most households.

Table 7 of the paper: France's households grouped at the U.S. income percentiles, so the population shares are the U.S. ones and each French group can stand in for its U.S. counterpart in the counterfactuals of section OA.3.4. The productivity profile is flatter than the U.S. one and the propensity to vote is nearly flat.

Table 8 of the paper: the UK's households at its own cuts, half-mean and mean income as for the U.S., which put most of them in the medium group. The propensity to vote rises with income, less steeply than in the U.S.

Table 9 of the paper: France's households regrouped at the UK's income percentiles, the groups imposed on the UK in section OA.3.5 and, with the design chosen, in section OA.4.3. Propensities to vote are observed at the U.S. cuts only, so each group carries the population-weighted average of the observed values it overlaps. At these cuts France's productivity spread is wider than the UK's (table 8 of the paper).

Under the vector- X_i calibration only X , whose level is the unit of hours, and the productivity spread change; θ , ω , the demography and the shared parameters are those of the common X . The spread widens in all three countries, the UK's now slightly above the U.S. one, and France's stays the narrowest.

Table OA.12: Calibration, the U.S., the UK and France (vector X_i calibration)

Parameter	U.S.	UK	France	Identified by
θ	0.74	0.56	1.00	Replacement rate dispersion
ω	1.45	1.16	1.42	$\tau^{US} = 14.4\%$, $\tau^{UK} = 11.9\%$, $\tau^{FR} = 21.3\%$
X	0.88	1.55	1.36	Average workweek
ν_{2020}	1.34	1.17	0.97	30-year gross population growth
η_H/η_L	3.73	3.87	2.41	Relative productivity, high to low income
β		0.76		30-year interest rate (U.S.)
ρ		1		Log-GHH preferences
ξ		0.30		Frisch elasticity of labor supply
α		0.30		Capital income share

Note: X is the population-weighted mean of X_i ; its level is the hours unit, pinned for France and the UK by targeting average hours relative to the U.S. rather than in levels. The lower panel is common to the three countries; β is calibrated for the U.S. and imposed on the other two. Vector- X_i calibration: X_i is identified from relative hours, which are data here, and the level of $\bar{h}$ is then not identified — only its ratio to the baseline is. β , ω , the tax rate and the savings rate are common to the two variants; what differs is anything defined through η or X .

Under the vector- X_i calibration the taste for leisure differs across the groups, identified from relative hours, and rises with income, and the productivities spread further apart than under the common X ; the population shares and the propensities to vote are data and do not change.

Table OA.13: Household heterogeneity – the U.S. (vector X_i calibration)

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	0.65	0.80	1.49	Hours worked.
η_i	0.54	0.95	2.00	Income distribution.
μ_i	0.47	0.63	0.77	Voting propensity.

Under the vector- X_i calibration the taste for leisure differs across the groups, identified from relative hours, and rises with income, and the productivities spread further apart than under the common X ; the population shares and the propensities to vote are data and do not change.

Under the vector- X_i calibration the taste for leisure differs across the groups, identified from relative hours, and rises with income, and the productivities spread further apart than under the common X ; the population shares and the propensities to vote are data and do not change.

Under the vector- X_i calibration the taste for leisure differs across the groups, identified from relative hours, and rises with income, and the productivities spread further apart than under the

Table OA.14: Household heterogeneity – France (vector X_i calibration)

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	1.27	1.34	1.59	Hours worked.
η_i	0.68	0.98	1.64	Income distribution.
μ_i	0.81	0.86	0.85	Voting propensity.

Table OA.15: Household heterogeneity – UK (vector X_i calibration)

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	1.08	1.48	2.97	Hours worked.
η_i	0.53	0.98	2.06	Income distribution.
μ_i	0.61	0.70	0.74	Voting propensity.

common X ; the population shares and the propensities to vote are data and do not change. At these cuts France's productivity spread is narrower than the UK's (table OA.15), where under the common X it is wider.

Table OA.16: Household heterogeneity – France, UK income groups (vector X_i calibration)

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	1.26	1.33	1.63	Hours worked.
η_i	0.57	0.93	2.01	Income distribution.
μ_i	0.81	0.83	0.85	Voting propensity.

The UK's households regrouped at the U.S. income percentiles, so the population shares are the U.S. ones. At these cuts the UK's productivity profile is much flatter than at its own (table 8 of the paper), the source of the low dispersion of relative incomes in the regrouped row of table 4 of the paper.

Under the vector- X_i calibration the taste for leisure differs across the groups, identified from relative hours, and rises with income, and the productivities spread further apart than under the common X ; the population shares and the propensities to vote are data and do not change.

Table OA.17: Household heterogeneity – UK, U.S. income groups

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	8.77	8.77	8.77	Average hours worked.
η_i	1.38	1.74	2.19	Income distribution.
μ_i	0.66	0.69	0.73	Voting propensity.

Table OA.18: Household heterogeneity – UK, U.S. income groups (vector X_i calibration)

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	1.41	1.70	1.99	Hours worked.
η_i	0.81	1.06	1.39	Income distribution.
μ_i	0.66	0.69	0.73	Voting propensity.

OA.3.2 The vector- X_i calibration

The paper identifies each income group's productivity and taste for leisure by taking the taste for leisure common across groups and fixing its level by the average workweek, so that relative hours are a prediction. The alternative takes the observed relative hours as data as well, which identifies one X_i per group and leaves the level of hours unidentified. The productivity distribution reaches every aggregate only through $\sum_i \gamma_i \eta_i^{1+\xi} / X_i^\xi$, which both calibrations normalize to one, so the calibration is block recursive: what depends on the design or on the demography is the same under both, exactly in the log case and, in the CRRA solves for the UK and France, to within one unit of the last printed digit, and only what is defined through η_i or X_i separately can move. The table compares every exhibit that exists under both calibrations, cell by cell, the Argentine exhibits of OA.2 included.

The cells that differ between the two calibrations, table by table

Table	Cells that differ	Rows in which they do
Calibration, Argentina	6 of 36	X, η_i, η_0, X_0
Pension system reform, year 2010	0 of 12	none
Pension system reform along the path, Argentina	0 of 42	none
Pension system reform, year 2010, function of ρ	0 of 68	none
The calibration across ρ , Argentina	48 of 128	0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0
Calibration, the U.S., the UK and France	6 of 37	$X, \eta_H / \eta_L$
Household heterogeneity – the U.S.	7 of 20	X_i, η_i
Household heterogeneity – France	7 of 20	X_i, η_i
Household heterogeneity – UK	7 of 20	X_i, η_i

The cells that differ between the two calibrations, table by table

Table	Cells that differ	Rows in which they do
Household heterogeneity – France, UK income groups	7 of 20	X_i, η_i
Household heterogeneity – UK, U.S. income groups	7 of 20	X_i, η_i
The calibration across ρ : the U.S., the UK and France	55 of 291	0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0
The effect of pension design (θ) in the U.S. – 2020	0 of 22	none
The effect of ageing in the U.S. – 2020	0 of 22	none
French income distribution, leisure preferences and voting patterns in the U.S.	7 of 24	Income distribution, Leisure preferences, All French characteristics
The effect of pension design (θ) across the IES, the U.S., 2020	0 of 45	none
The effect of ageing across the IES, the U.S., 2020	0 of 45	none
French characteristics imposed on the U.S. across the IES, 2020	21 of 75	Income distribution, Leisure preferences, All French characteristics
French income distribution, leisure preferences and voting patterns in the UK	7 of 24	Income distribution, Leisure preferences, All French characteristics
French characteristics imposed on the UK across the IES, 2020	23 of 75	Baseline, Income distribution, Leisure preferences, Voting patterns, All French characteristics
The effect of pension design (θ) across the IES, the UK, 2020	3 of 45	Baseline, $\theta = 0$
The effect of ageing across the IES, the UK, 2020	4 of 45	Baseline, Mild ageing, Acute ageing

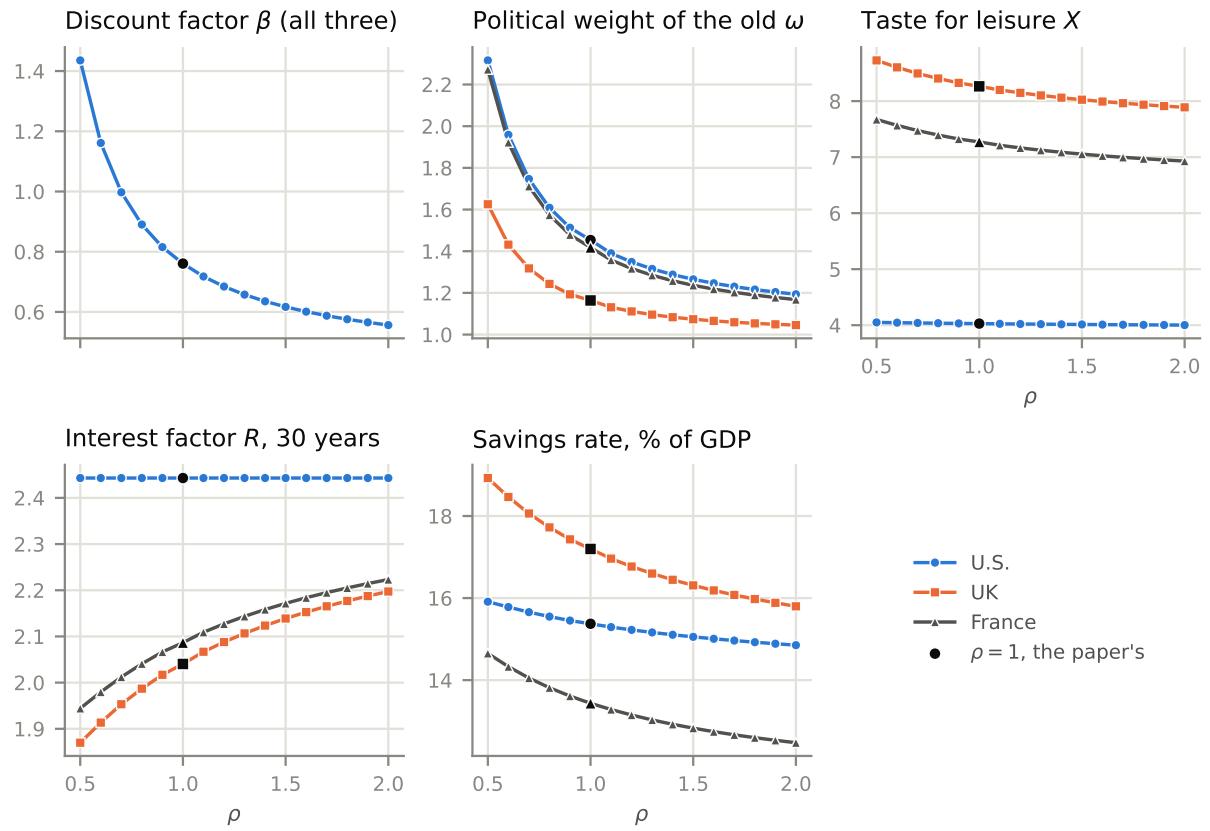
OA.3.3 The calibration across the intertemporal elasticity

The figure and the table print the parameters solved jointly with the equilibrium path at every IES of the grid, for each country. Sections 6 and 7 explain the CRRA results by how the calibration absorbs a higher IES; these are the numbers behind that explanation. Matching the U.S. targets at an IES of 0.6 or below takes a thirty-year discount factor above one, which the UK and France inherit.

One line per country; the discount factor is the U.S. one in all three. The political weight of the old falls as the IES rises in every country, since the young resist taxation less, and keeps its ranking, highest in the U.S. and lowest in the UK; the savings rate falls in every country, and the interest factor of the UK and France, a prediction, rises toward the U.S. target.

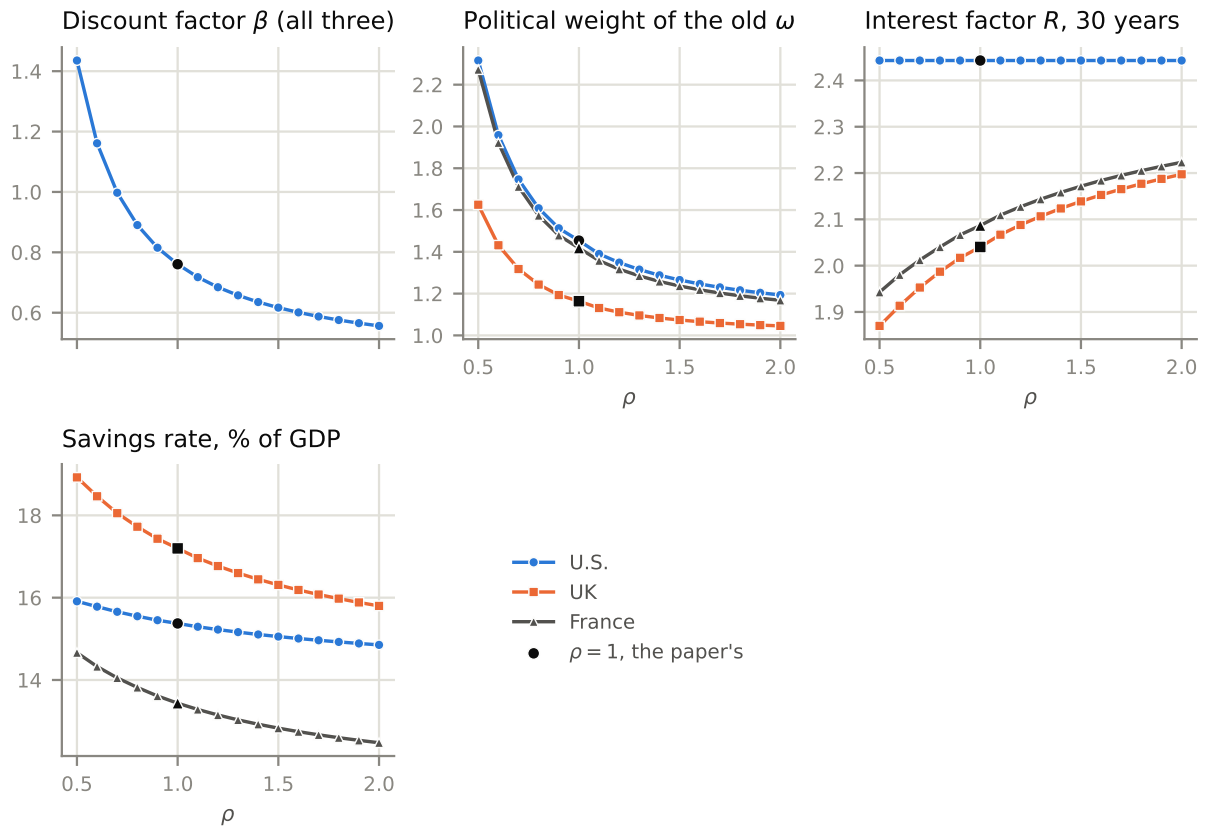
Under the vector- X_i calibration there is no common taste for leisure to draw, and every other panel is that of the common X up to the last printed digit.

Figure OA.6: The calibrations of the U.S., the UK and France across the intertemporal elasticity



Note: The parameters solved jointly with the equilibrium path at each IES of the grid, against the same targets, with β calibrated for the U.S. and imposed on the UK and France; the table of this section prints their values, and $\rho = 1$ is the paper's calibration.

Figure OA.7: The calibrations of the U.S., the UK and France across the intertemporal elasticity, vector- X_i calibration



Note: The parameters solved jointly with the equilibrium path at each IES of the grid, against the same targets, with β calibrated for the U.S. and imposed on the UK and France; the table of this section prints their values, and $\rho = 1$ is the paper's calibration.

Table OA.19: The calibration across ρ : the U.S., the UK and France

CRRA (ρ)	β	ω	X	R	Savings rate
<i>U.S.</i>					
0.5	1.435	2.316	4.05	2.443	15.91%
0.6	1.161	1.959	4.04	2.443	15.78%
0.7	0.997	1.747	4.04	2.443	15.66%
0.8	0.890	1.609	4.04	2.443	15.55%
0.9	0.815	1.513	4.03	2.443	15.45%
1.0	0.761	1.454	4.03	2.443	15.37%
1.1	0.718	1.390	4.02	2.443	15.29%
1.2	0.684	1.348	4.02	2.443	15.22%
1.3	0.658	1.315	4.02	2.443	15.16%
1.4	0.635	1.288	4.02	2.443	15.11%
1.5	0.617	1.265	4.01	2.443	15.06%
1.6	0.601	1.246	4.01	2.443	15.01%
1.7	0.587	1.230	4.01	2.443	14.96%
1.8	0.576	1.216	4.01	2.443	14.92%
1.9	0.565	1.204	4.00	2.443	14.89%
2.0	0.556	1.193	4.00	2.443	14.85%
<i>UK</i>					
0.5	1.435	1.625	8.73	1.870	18.93%
0.6	1.161	1.432	8.60	1.913	18.46%
0.7	0.997	1.318	8.49	1.953	18.06%
0.8	0.890	1.243	8.40	1.987	17.72%
0.9	0.815	1.193	8.33	2.017	17.43%
1.0	0.761	1.164	8.26	2.040	17.20%
1.1	0.718	1.131	8.20	2.067	16.96%
1.2	0.684	1.111	8.15	2.088	16.77%
1.3	0.658	1.096	8.10	2.107	16.60%
1.4	0.635	1.083	8.06	2.124	16.45%
1.5	0.617	1.074	8.02	2.139	16.31%
1.6	0.601	1.066	7.99	2.153	16.19%
1.7	0.587	1.059	7.96	2.165	16.08%
1.8	0.576	1.054	7.94	2.177	15.98%
1.9	0.565	1.049	7.91	2.187	15.88%
2.0	0.556	1.046	7.89	2.197	15.80%
<i>France</i>					
0.5	1.435	2.272	7.67	1.944	14.65%
0.6	1.161	1.922	7.57	1.980	14.33%
0.7	0.997	1.712	7.47	2.012	14.05%
0.8	0.890	1.574	7.39	2.041	13.81%
0.9	0.815	1.479	7.32	2.066	13.61%
1.0	0.761	1.418	7.27	2.087	13.44%

Table OA.20: The calibration across ρ : the U.S., the UK and France (vector X_i calibration)

CRRA (ρ)	β	ω	X	R	Savings rate
<i>U.S.</i>					
0.5	1.435	2.316	—	2.443	15.91%
0.6	1.161	1.959	—	2.443	15.78%
0.7	0.997	1.747	—	2.443	15.66%
0.8	0.890	1.609	—	2.443	15.55%
0.9	0.815	1.513	—	2.443	15.45%
1.0	0.761	1.454	—	2.443	15.37%
1.1	0.718	1.390	—	2.443	15.29%
1.2	0.684	1.348	—	2.443	15.22%
1.3	0.658	1.315	—	2.443	15.16%
1.4	0.635	1.288	—	2.443	15.11%
1.5	0.617	1.265	—	2.443	15.06%
1.6	0.601	1.246	—	2.443	15.01%
1.7	0.587	1.230	—	2.443	14.96%
1.8	0.576	1.216	—	2.443	14.92%
1.9	0.565	1.204	—	2.443	14.89%
2.0	0.556	1.193	—	2.443	14.85%
<i>UK</i>					
0.5	1.435	1.625	—	1.870	18.92%
0.6	1.161	1.432	—	1.913	18.46%
0.7	0.997	1.318	—	1.953	18.05%
0.8	0.890	1.243	—	1.987	17.72%
0.9	0.815	1.193	—	2.017	17.43%
1.0	0.761	1.164	—	2.040	17.20%
1.1	0.718	1.131	—	2.067	16.96%
1.2	0.684	1.111	—	2.088	16.77%
1.3	0.658	1.096	—	2.107	16.60%
1.4	0.635	1.083	—	2.123	16.45%
1.5	0.617	1.074	—	2.139	16.31%
1.6	0.601	1.066	—	2.153	16.19%
1.7	0.587	1.059	—	2.165	16.08%
1.8	0.576	1.054	—	2.177	15.98%
1.9	0.565	1.049	—	2.187	15.88%
2.0	0.556	1.046	—	2.197	15.80%
<i>France</i>					
0.5	1.435	2.272	—	1.943	14.66%
0.6	1.161	1.922	—	1.980	14.33%
0.7	0.997	1.712	—	2.012	14.05%
0.8	0.890	1.574	—	2.040	13.82%
0.9	0.815	1.479	—	2.066	13.61%

OA.3.4 Counterfactuals on the U.S.

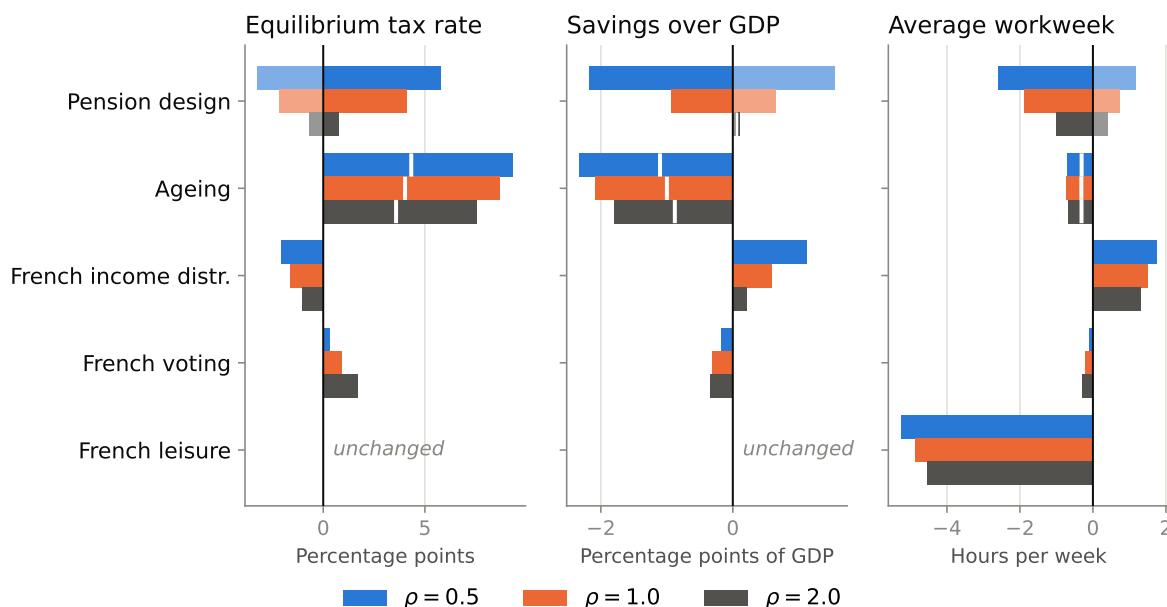
The paper's overview figure of section 6 collects the effect of each counterfactual on the tax rate, the savings rate and the workweek at three values of the IES. Each counterfactual is a separate equilibrium path read in 2020: one characteristic of the U.S. calibration is changed over the whole horizon and every other parameter, the political weight of the old included, stays at its U.S. value, so a row describes an economy that has always had that characteristic. The tables print the effects: the pension design and ageing at the paper's benchmark, with the economic-equilibrium effect, in which the tax rate is held at its baseline path, beside the full one; France's income distribution, leisure preferences and voting patterns one at a time and all at once, next to France's own calibrated path; and the same counterfactuals at each IES.

Beside these, the site shows the paper's figure 6.1 (section 6), which this edition does not reprint.

Figure 6.1 of the paper: one row per characteristic changed in the U.S. calibration, at three values of the IES. The design and ageing dominate the tax panel at an IES of one or below, while at an IES of two France's voting patterns move the tax rate by more than the whole range of the design. France's leisure preferences dominate the workweek panel and leave the tax and savings rates essentially unchanged.

Only the rows that impose France's productivities or its taste for leisure differ from the paper's figure: France's income distribution lowers the tax rate and lengthens the workweek by more at every IES, and its leisure preferences shorten the workweek a little less. The rows for the design, ageing and voting patterns are the paper's.

Figure OA.8: The effect of various counterfactuals in the U.S., vector- X_i calibration



Note: The paper's figure under the vector- X_i calibration (section OA.3.2); its baseline levels are the baseline rows of the tables of this section.

The design at its two corners against the observed one, at an IES of one. Moving to flat benefits raises the tax rate and lowers the savings rate and the workweek; with the tax held at its baseline

path it raises the savings rate instead, so the tax response is what lowers it, while for hours the two effects point the same way. Fully earnings-related benefits reverse every sign.

Table OA.21: The effect of pension design (θ) in the U.S. – 2020

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
$\theta = 0.74$	14.43%	15.37%	39.39
<i>Full effect:</i>			
$\theta = 0$	18.52%	−0.93 p.p.	37.52
$\theta = 1$	12.28%	+0.65 p.p.	40.13
<i>Economic equilibrium effect:</i>			
$\theta = 0$	14.43%	+0.41 p.p.	38.54
$\theta = 1$	14.43%	−0.14 p.p.	39.67

Note: $\rho = 1.0$. The economic-equilibrium rows hold τ at the baseline path, so they isolate the response of savings and hours to θ alone; the full rows re-optimize τ politically. The savings rate is savings relative to GDP; the baseline row reports its level and every other row the change against that baseline in percentage points.

Under the vector- X_i calibration this table is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so what the design and the demography move does not depend on how the taste for leisure is split across the income groups.

Table OA.22: The effect of pension design (θ) in the U.S. – 2020 (vector X_i calibration)

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
$\theta = 0.74$	14.43%	15.37%	39.39
<i>Full effect:</i>			
$\theta = 0$	18.52%	−0.93 p.p.	37.52
$\theta = 1$	12.28%	+0.65 p.p.	40.13
<i>Economic equilibrium effect:</i>			
$\theta = 0$	14.43%	+0.41 p.p.	38.54
$\theta = 1$	14.43%	−0.14 p.p.	39.67

Note: $\rho = 1.0$. The economic-equilibrium rows hold τ at the baseline path, so they isolate the response of savings and hours to θ alone; the full rows re-optimize τ politically. The savings rate is savings relative to GDP; the baseline row reports its level and every other row the change against that baseline in percentage points. Vector- X_i calibration: see the note to table OA.12.

The two ageing scenarios at an IES of one. Ageing raises the tax rate and lowers the savings rate and the workweek, acute ageing by more than mild; with the tax held at its baseline path the savings rate does not move and the workweek lengthens, so both falls come from the tax response.

Table OA.23: The effect of ageing in the U.S. – 2020

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	14.43%	15.37%	39.39
<i>Full effect:</i>			
Mild ageing ^a	18.45%	−1.00 p.p.	39.08
Acute ageing ^b	23.11%	−2.08 p.p.	38.66
<i>Economic equilibrium effect:</i>			
Mild ageing ^a	14.43%	0.00 p.p.	40.12
Acute ageing ^b	14.43%	0.00 p.p.	40.99

Note: Each scenario is a separate equilibrium path. The savings rate is savings relative to GDP; the baseline row reports its level and every other row the change against that baseline in percentage points.

^a The “mild ageing” scenario refers to the case with ν_t set at $(1 + \nu_t^{base})/2$ throughout.

^b The “acute ageing” scenario refers to $\nu_t = 1$ throughout.

Under the vector- X_i calibration this table is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so what the design and the demography move does not depend on how the taste for leisure is split across the income groups.

Table OA.24: The effect of ageing in the U.S. – 2020 (vector X_i calibration)

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	14.43%	15.37%	39.39
<i>Full effect:</i>			
Mild ageing ^a	18.45%	−1.00 p.p.	39.08
Acute ageing ^b	23.11%	−2.08 p.p.	38.66
<i>Economic equilibrium effect:</i>			
Mild ageing ^a	14.43%	0.00 p.p.	40.12
Acute ageing ^b	14.43%	0.00 p.p.	40.99

Note: Each scenario is a separate equilibrium path. The savings rate is savings relative to GDP; the baseline row reports its level and every other row the change against that baseline in percentage points.

^a The “mild ageing” scenario refers to the case with ν_t set at $(1 + \nu_t^{base})/2$ throughout.

^b The “acute ageing” scenario refers to $\nu_t = 1$ throughout. Vector- X_i calibration: see the note to table OA.12.

France's characteristics on the U.S. at an IES of one. France's income distribution lowers the tax rate and raises the savings rate and the workweek, its voting patterns raise the tax rate, and its leisure preferences shorten the workweek and move nothing else. All three together bring the workweek close to France's but move the tax rate away from it.

Table OA.25: French income distribution, leisure preferences and voting patterns in the U.S.

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	14.43%	15.37%	39.39
Income distribution	13.21%	+0.44 p.p.	40.36
Leisure preferences	14.43%	0.00 p.p.	34.37
Voting patterns	15.32%	−0.32 p.p.	39.19
All French characteristics	13.85%	+0.21 p.p.	35.09
France (own calibration)	21.29%	−1.94 p.p.	35.24

Note: $\rho = 1.0$, full effect. Each row is a separate equilibrium path: the borrowed characteristics hold throughout and the economy starts from its own steady state, so the row describes a country that has always had this mix rather than the U.S. hit by a surprise in 2020. Income distribution replaces η_i with France's while holding X_i and θ at their U.S. values, so it is a change in inequality alone; pension design is the separate counterfactual of table OA.21. Leisure preferences keeps the U.S. η_i and rescales every X_i to France's population-weighted mean X , scaled as France's η_i are in the income row; it is a pure rescaling of the hours unit and moves hours and nothing else. All French characteristics imposes France's η_i , level of X_i and voting weights μ_i at once. The last row is France's own calibrated path, its savings rate reported as the distance from the U.S. baseline.

Under the vector- X_i calibration only the rows that impose France's productivities or its taste for leisure differ: France's income distribution lowers the tax rate and raises the savings rate by more, alone and with the other characteristics, and the workweek is longer in each of these rows. Every sign is that of the common X , and the voting row and France's own path are unchanged.

The design at its two corners at each IES. Flat benefits raise the tax rate and shorten the workweek at every IES and fully earnings-related ones do the reverse, both by less the higher the IES, since the calibration then needs a lower political weight of the old and the difference between rich and poor retirees counts for less. Flat benefits lower the savings rate at $\rho = 0.5$ and 1 but raise it a little at $\rho = 2$, where both corners raise it.

Under the vector- X_i calibration this table is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so what the design and the demography move does not depend on how the taste for leisure is split across the income groups.

The two ageing scenarios at each IES. Ageing raises the tax rate and lowers the savings rate and the workweek at every IES, acute ageing by more than mild, and the tax response falls only a little as the IES rises, where that of the design falls steeply (table OA.27).

Under the vector- X_i calibration this table is identical to the common- X one to the last printed digit, as the comparison in section OA.3.2 records: the calibration is block recursive, so what the design and the demography move does not depend on how the taste for leisure is split across the income groups.

France's characteristics on the U.S. at each IES. The income distribution lowers the tax rate at every IES and the voting patterns raise it, the first by less and the second by more the higher the

Table OA.26: French income distribution, leisure preferences and voting patterns in the U.S. (vector X_i calibration)

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	14.43%	15.37%	39.39
Income distribution	12.80%	+0.59 p.p.	40.89
Leisure preferences	14.43%	0.00 p.p.	34.52
Voting patterns	15.32%	−0.32 p.p.	39.19
All French characteristics	13.34%	+0.39 p.p.	35.72
France (own calibration)	21.29%	−1.94 p.p.	35.24

Note: $\rho = 1.0$, full effect. Each row is a separate equilibrium path: the borrowed characteristics hold throughout and the economy starts from its own steady state, so the row describes a country that has always had this mix rather than the U.S. hit by a surprise in 2020. Income distribution replaces η_i with France's while holding X_i and θ at their U.S. values, so it is a change in inequality alone; pension design is the separate counterfactual of table OA.21. Leisure preferences keeps the U.S. η_i and rescales every X_i to France's population-weighted mean X , scaled as France's η_i are in the income row; it is a pure rescaling of the hours unit and moves hours and nothing else. All French characteristics imposes France's η_i , level of X_i and voting weights μ_i at once. The last row is France's own calibrated path, its savings rate reported as the distance from the U.S. baseline. Vector- X_i calibration: see the note to table OA.12.

Table OA.27: The effect of pension design (θ) across the IES, the U.S., 2020

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	14.43%	15.91%	39.39
	1.0	14.43%	15.37%	39.39
	2.0	14.43%	14.85%	39.39
$\theta = 0$	0.5	20.23%	−2.18 p.p.	36.80
	1.0	18.52%	−0.93 p.p.	37.52
	2.0	15.21%	+0.10 p.p.	38.38
$\theta = 1$	0.5	11.21%	+1.54 p.p.	40.55
	1.0	12.28%	+0.65 p.p.	40.13
	2.0	13.72%	+0.06 p.p.	39.79

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the U.S. calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points.

Table OA.28: The effect of pension design (θ) across the IES, the U.S., 2020 (vector X_i calibration)

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	14.43%	15.91%	39.39
	1.0	14.43%	15.37%	39.39
	2.0	14.43%	14.85%	39.39
$\theta = 0$	0.5	20.23%	-2.18 p.p.	36.80
	1.0	18.52%	-0.93 p.p.	37.52
	2.0	15.21%	+0.10 p.p.	38.38
$\theta = 1$	0.5	11.21%	+1.54 p.p.	40.55
	1.0	12.28%	+0.65 p.p.	40.13
	2.0	13.72%	+0.06 p.p.	39.79

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the U.S. calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. Vector- X_i calibration: see the note to table OA.12.

IES, and the leisure preferences move essentially the workweek alone. With all three together the tax rate falls at $\rho = 0.5$ and 1 and rises at $\rho = 2$, where the voting patterns prevail.

Under the vector- X_i calibration only the rows that impose France's productivities or its taste for leisure differ: the income distribution lowers the tax rate by more at every IES, and with all three characteristics together the tax rate falls at every IES, only marginally at $\rho = 2$, where under the common X it rises. The voting rows are those of the common X .

Table OA.29: The effect of ageing across the IES, the U.S., 2020

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	14.43%	15.91%	39.39
	1.0	14.43%	15.37%	39.39
	2.0	14.43%	14.85%	39.39
Mild ageing	0.5	18.76%	−1.10 p.p.	39.08
	1.0	18.45%	−1.00 p.p.	39.08
	2.0	18.01%	−0.88 p.p.	39.08
Acute ageing	0.5	23.75%	−2.32 p.p.	38.68
	1.0	23.11%	−2.08 p.p.	38.66
	2.0	21.99%	−1.79 p.p.	38.72

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the U.S. calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points.

Table OA.30: The effect of ageing across the IES, the U.S., 2020 (vector X_i calibration)

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	14.43%	15.91%	39.39
	1.0	14.43%	15.37%	39.39
	2.0	14.43%	14.85%	39.39
Mild ageing	0.5	18.76%	−1.10 p.p.	39.08
	1.0	18.45%	−1.00 p.p.	39.08
	2.0	18.01%	−0.88 p.p.	39.08
Acute ageing	0.5	23.75%	−2.32 p.p.	38.68
	1.0	23.11%	−2.08 p.p.	38.66
	2.0	21.99%	−1.79 p.p.	38.72

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the U.S. calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. Vector- X_i calibration: see the note to table OA.12.

Table OA.31: French characteristics imposed on the U.S. across the IES, 2020

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	14.43%	15.91%	39.39
	1.0	14.43%	15.37%	39.39
	2.0	14.43%	14.85%	39.39
Income distribution	0.5	12.88%	+0.83 p.p.	40.53
	1.0	13.21%	+0.44 p.p.	40.36
	2.0	13.65%	+0.16 p.p.	40.22
Leisure preferences	0.5	14.41%	0.00 p.p.	33.99
	1.0	14.43%	0.00 p.p.	34.37
	2.0	14.43%	0.00 p.p.	34.71
Voting patterns	0.5	14.75%	−0.17 p.p.	39.30
	1.0	15.32%	−0.32 p.p.	39.19
	2.0	16.12%	−0.34 p.p.	39.11
All French characteristics	0.5	13.15%	+0.67 p.p.	34.91
	1.0	13.85%	+0.21 p.p.	35.09
	2.0	14.86%	−0.09 p.p.	35.25

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the U.S. calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points.

Table OA.32: French characteristics imposed on the U.S. across the IES, 2020 (vector X_i calibration)

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	14.43%	15.91%	39.39
	1.0	14.43%	15.37%	39.39
	2.0	14.43%	14.85%	39.39
Income distribution	0.5	12.36%	+1.11 p.p.	41.12
	1.0	12.80%	+0.59 p.p.	40.89
	2.0	13.41%	+0.21 p.p.	40.70
Leisure preferences	0.5	14.41%	0.00 p.p.	34.14
	1.0	14.43%	0.00 p.p.	34.52
	2.0	14.43%	0.00 p.p.	34.85
Voting patterns	0.5	14.75%	−0.17 p.p.	39.30
	1.0	15.32%	−0.32 p.p.	39.19
	2.0	16.12%	−0.34 p.p.	39.11
All French characteristics	0.5	12.61%	+0.97 p.p.	35.57
	1.0	13.34%	+0.39 p.p.	35.72
	2.0	14.41%	0.00 p.p.	35.85

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the U.S. calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. Vector- X_i calibration: see the note to table OA.12.

OA.3.5 The UK as host

The counterfactuals of section 6 on the UK's own calibration, each a separate equilibrium path with one characteristic changed over the whole horizon and every other parameter at the UK's value: France's characteristics, with France's households regrouped at the UK's income percentiles, and the pension design and ageing at each IES. The figure sets the two hosts side by side for France's characteristics; the tables print the UK's numbers. Two differences from the U.S. stand out. France's income distribution raises the UK's tax rate where it lowers the U.S. one, because at the UK's income cuts France is the more unequal of the two countries (OA.3.1) and equilibrium taxes rise with inequality; and France's taste for leisure lengthens the UK's workweek where it shortens the U.S. one, since France's average taste for leisure lies between the two hosts'. France's voting profile raises the tax rate on both hosts, by more at a higher IES. The income-distribution row is also the one that depends on the calibration of the taste for leisure: under the vector- X_i calibration it lowers the UK's tax rate, because that calibration reads part of the observed income dispersion off relative hours, which are much flatter in the UK than in France.

France's characteristics on the two hosts, the U.S. in the upper row of panels and the UK in the lower, on shared axes so that the hosts compare by position; the last row imposes all three at once. Together they raise the UK's tax rate at every IES, while on the U.S. they lower it at $\rho = 0.5$ and 1 and raise it at $\rho = 2$.

The same comparison under the vector- X_i calibration, which changes the rows that impose France's productivities or its taste for leisure. France's income distribution lowers the tax rate on both hosts at every IES, and all three characteristics together lower it on both at $\rho = 0.5$ and 1, while at $\rho = 2$ they raise the UK's a little and leave the U.S. one essentially unchanged.

France's characteristics on the UK at an IES of one, laid out as the U.S. table OA.25. France's income distribution and voting patterns each move the UK's tax rate by less than the U.S. one, and with all three characteristics together the UK's tax rate rises toward France's, where the U.S. one moves away from it.

Under the vector- X_i calibration the rows that impose France's productivities or its taste for leisure differ: France's income distribution lowers the UK's tax rate and raises its savings rate and workweek, the reverse of every common- X sign, and with all three characteristics together the UK's tax rate falls a little. The voting row and France's own path are unchanged.

France's characteristics on the UK at each IES. Each moves the UK's savings rate by little and the leisure preferences move the workweek alone; the income distribution and the voting patterns both raise the tax rate, by more the higher the IES, and so do all three together.

Under the vector- X_i calibration France's income distribution lowers the UK's tax rate at every IES, by less the higher the IES, and all three characteristics together lower it at $\rho = 0.5$ and 1 and raise it at $\rho = 2$; the voting rows are essentially those of the common X .

The design at its two corners on the UK at each IES. As on the U.S. (table OA.27), flat benefits raise the tax rate and shorten the workweek at every IES and fully earnings-related ones do the reverse, both by less the higher the IES; unlike on the U.S., flat benefits lower the savings rate at every IES.

Under the vector- X_i calibration this table differs from the common- X one only in the last printed digit of a few cells at the lowest IES, as the comparison in section OA.3.2 records: the

Figure OA.9: French characteristics in the U.S. and the UK



Note: Every panel is the deviation from the host's own baseline at the same ρ , whose levels are the baseline rows of the tables of this section. France's income groups are cut at the host's income percentiles. The savings rate is savings relative to GDP.

Figure OA.10: French characteristics in the U.S. and the UK, vector- X_i calibration



Note: Every panel is the deviation from the host's own baseline at the same ρ , whose levels are the baseline rows of the tables of this section. France's income groups are cut at the host's income percentiles. The savings rate is savings relative to GDP.

Table OA.33: French income distribution, leisure preferences and voting patterns in the UK

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	11.86%	17.20%	34.73
Income distribution	12.15%	−0.12 p.p.	34.55
Leisure preferences	11.86%	0.00 p.p.	35.85
Voting patterns	12.19%	−0.13 p.p.	34.66
All French characteristics	12.46%	−0.24 p.p.	35.61
France (own calibration)	21.29%	−3.76 p.p.	35.24

Note: $\rho = 1.0$, full effect. Each row is a separate equilibrium path: the borrowed characteristics hold throughout and the economy starts from its own steady state, so the row describes a country that has always had this mix rather than the UK hit by a surprise in 2020. Income distribution replaces η_i with France's while holding X_i and θ at their UK values, so it is a change in inequality alone. Leisure preferences keeps the UK η_i and rescales every X_i to France's population-weighted mean X , scaled as France's η_i are in the income row; it is a pure rescaling of the hours unit and moves hours and nothing else. All French characteristics imposes France's η_i , level of X_i and voting weights μ_i at once. The last row is France's own calibrated path, its savings rate reported as the distance from the UK baseline. France's income groups are cut at the UK's own income percentiles here (table 9 of the paper).

Table OA.34: French income distribution, leisure preferences and voting patterns in the UK (vector X_i calibration)

Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	11.86%	17.20%	34.73
Income distribution	11.40%	+0.19 p.p.	34.93
Leisure preferences	11.86%	0.00 p.p.	35.99
Voting patterns	12.19%	−0.13 p.p.	34.66
All French characteristics	11.66%	+0.08 p.p.	36.15
France (own calibration)	21.29%	−3.76 p.p.	35.24

Note: $\rho = 1.0$, full effect. Each row is a separate equilibrium path: the borrowed characteristics hold throughout and the economy starts from its own steady state, so the row describes a country that has always had this mix rather than the UK hit by a surprise in 2020. Income distribution replaces η_i with France's while holding X_i and θ at their UK values, so it is a change in inequality alone. Leisure preferences keeps the UK η_i and rescales every X_i to France's population-weighted mean X , scaled as France's η_i are in the income row; it is a pure rescaling of the hours unit and moves hours and nothing else. All French characteristics imposes France's η_i , level of X_i and voting weights μ_i at once. The last row is France's own calibrated path, its savings rate reported as the distance from the UK baseline. France's income groups are cut at the UK's own income percentiles here (table 9 of the paper). Vector- X_i calibration: see the note to table OA.12.

calibration is block recursive, so what the design and the demography move does not depend on how the taste for leisure is split across the income groups.

Table OA.35: French characteristics imposed on the UK across the IES, 2020

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	11.86%	18.93%	34.73
	1.0	11.86%	17.20%	34.73
	2.0	11.86%	15.80%	34.73
Income distribution	0.5	11.94%	−0.05 p.p.	34.59
	1.0	12.15%	−0.12 p.p.	34.55
	2.0	12.23%	−0.09 p.p.	34.55
Leisure preferences	0.5	11.86%	0.00 p.p.	35.86
	1.0	11.86%	0.00 p.p.	35.85
	2.0	11.86%	0.00 p.p.	35.87
Voting patterns	0.5	12.08%	−0.14 p.p.	34.67
	1.0	12.19%	−0.13 p.p.	34.66
	2.0	12.39%	−0.12 p.p.	34.65
All French characteristics	0.5	12.14%	−0.18 p.p.	35.66
	1.0	12.46%	−0.24 p.p.	35.61
	2.0	12.76%	−0.20 p.p.	35.60

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the UK calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points.

The two ageing scenarios on the UK at each IES. As on the U.S., ageing raises the tax rate and lowers the savings rate and the workweek at every IES, acute ageing by more than mild, and the tax response falls only a little as the IES rises; every effect is smaller than on the U.S. (table OA.29).

Under the vector- X_i calibration this table differs from the common- X one only in the last printed digit of a few cells at the lowest IES, as the comparison in section OA.3.2 records: the calibration is block recursive, so what the design and the demography move does not depend on how the taste for leisure is split across the income groups.

Table OA.36: French characteristics imposed on the UK across the IES, 2020 (vector X_i calibration)

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	11.86%	18.92%	34.73
	1.0	11.86%	17.20%	34.73
	2.0	11.86%	15.80%	34.73
Income distribution	0.5	11.06%	+0.49 p.p.	35.03
	1.0	11.40%	+0.19 p.p.	34.93
	2.0	11.56%	+0.07 p.p.	34.89
Leisure preferences	0.5	11.86%	0.00 p.p.	35.99
	1.0	11.86%	0.00 p.p.	35.99
	2.0	11.86%	0.00 p.p.	36.00
Voting patterns	0.5	12.09%	−0.14 p.p.	34.67
	1.0	12.19%	−0.13 p.p.	34.66
	2.0	12.39%	−0.12 p.p.	34.65
All French characteristics	0.5	11.24%	+0.38 p.p.	36.26
	1.0	11.66%	+0.08 p.p.	36.15
	2.0	12.01%	−0.04 p.p.	36.10

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the UK calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. Vector- X_i calibration: see the note to table OA.12.

Table OA.37: The effect of pension design (θ) across the IES, the UK, 2020

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	11.86%	18.93%	34.73
	1.0	11.86%	17.20%	34.73
	2.0	11.86%	15.80%	34.73
$\theta = 0$	0.5	14.77%	-1.24 p.p.	33.56
	1.0	14.03%	-0.58 p.p.	33.83
	2.0	13.15%	-0.16 p.p.	34.09
$\theta = 1$	0.5	8.50%	+1.85 p.p.	35.76
	1.0	9.60%	+0.76 p.p.	35.44
	2.0	10.39%	+0.25 p.p.	35.23

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the UK calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points.

Table OA.38: The effect of pension design (θ) across the IES, the UK, 2020 (vector X_i calibration)

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	11.86%	18.92%	34.73
	1.0	11.86%	17.20%	34.73
	2.0	11.86%	15.80%	34.73
$\theta = 0$	0.5	14.78%	-1.25 p.p.	33.56
	1.0	14.03%	-0.58 p.p.	33.83
	2.0	13.15%	-0.16 p.p.	34.09
$\theta = 1$	0.5	8.50%	+1.85 p.p.	35.76
	1.0	9.60%	+0.76 p.p.	35.44
	2.0	10.39%	+0.25 p.p.	35.23

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the UK calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. Vector- X_i calibration: see the note to table OA.12.

Table OA.39: The effect of ageing across the IES, the UK, 2020

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	11.86%	18.93%	34.73
	1.0	11.86%	17.20%	34.73
	2.0	11.86%	15.80%	34.73
Mild ageing	0.5	14.26%	−0.77 p.p.	34.67
	1.0	14.12%	−0.77 p.p.	34.63
	2.0	13.96%	−0.77 p.p.	34.60
Acute ageing	0.5	16.85%	−1.57 p.p.	34.60
	1.0	16.59%	−1.57 p.p.	34.52
	2.0	16.19%	−1.54 p.p.	34.47

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the UK calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points.

Table OA.40: The effect of ageing across the IES, the UK, 2020 (vector X_i calibration)

Scenario	CRRA (ρ)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	11.86%	18.92%	34.73
	1.0	11.86%	17.20%	34.73
	2.0	11.86%	15.80%	34.73
Mild ageing	0.5	14.26%	−0.77 p.p.	34.66
	1.0	14.12%	−0.77 p.p.	34.63
	2.0	13.96%	−0.77 p.p.	34.60
Acute ageing	0.5	16.86%	−1.57 p.p.	34.59
	1.0	16.59%	−1.57 p.p.	34.52
	2.0	16.19%	−1.54 p.p.	34.47

Note: Every ρ is separately calibrated. Each counterfactual is a separate equilibrium path on which the changed characteristic holds throughout, every other parameter at its value in the UK calibration. The tax rate and the workweek are levels. The savings rate is savings relative to GDP: the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. Vector- X_i calibration: see the note to table OA.12.

OA.4 The endogenous pension design

Section 7 of the paper lets the electorate choose the design θ as well as the tax rate, under a deadweight cost of redistribution whose parameter λ is calibrated so that the U.S. electorate re-elects its observed design. This chapter holds the tables behind section 7, the same counterfactuals with the UK as host, the design along the baseline path, the timing of the choice and an alternative cost specification.

OA.4.1 The cost and the cross-country test

The calibrated cost at each intertemporal elasticity of substitution (IES), and the paper's table of section 7.2: the designs the UK and France choose under the U.S. parameter, with the UK's own parameter; France, at the corner, has none.

Beside these, the site shows the paper's table 4 (section 7), which this edition does not reprint.

Table 4 of the paper: the test of the cost parameter calibrated on the U.S. alone, at an IES of one. Under the U.S. parameter the chosen designs keep the observed ranking of the UK, the U.S. and France, compressed toward the U.S. value; the UK's own parameter lies below the U.S. one, and regrouping the UK at the U.S. income percentiles moves its chosen design only slightly.

Table OA.41: one row per IES, each with the cost parameter at which the U.S. electorate re-elects its observed design in 2020. The parameter, and with it the share of revenue lost at the observed design and at flat benefits, falls steeply as the IES rises, since a higher elasticity strengthens the forward-looking channels that hold the choice away from the Beveridgean corner.

Table OA.41: The calibrated cost of redistribution

CRRA (ρ)	λ	θ^*	$f(\theta^*)$	$f(0)$	$\tilde{V}$
0.5	18.267	0.738	0.961	0.563	0.436
1.0	8.643	0.738	0.982	0.762	0.436
2.0	1.724	0.738	0.996	0.947	0.436

Note: $f(\theta, \tau) = \exp\{-\frac{1}{2}\lambda\tau\tilde{V}(1-\theta)^2\}$ with $\tilde{V} = \sum_i \gamma_i (y_i - 1)^2 / y_i$ the dispersion of relative labor incomes; $f(\theta^*)$ and $f(0)$ are read at the 2020 tax rate, so $1 - f$ is the share of revenue lost at the observed design and at a flat benefit; λ is calibrated so the design in force in 2020 is the observed one, with (β, ω) recalibrated at each trial value. The cost is proportional, so it cancels from the replacement-rate ratio and θ^* is the data's own. Without the cost the choice would be in the corner $\theta = 0$ for every ρ .

OA.4.2 Counterfactuals on the U.S.

Each counterfactual is a separate equilibrium path read in 2020, with one characteristic changed over the whole horizon, every other parameter at its U.S. value and the political choice binding from the first period. Each is read twice, with the design pinned at the U.S. value and with the design chosen, and both readings carry the cost, so the distance between them is the political response. The figure's last row, France's income distribution and voting patterns together, shares its design, tax rate and savings rate with the table's row that imposes all French characteristics, since France's level of X moves hours and nothing else. The paper's table is the first, at an IES of one; the figure and the other tables hold every IES. In the figure an open marker is the pinned reading and a filled one the chosen.

Beside these, the site shows the paper's table 5 (section 7), which this edition does not reprint.

Table 5 of the paper: the paper's reading at an IES of one, the rows of the summary table below without the composite. Acute ageing makes the chosen design more Bismarckian and France's voting patterns less so, by about as much; France's income distribution moves it little, and in every row the tax rate with the design chosen lies within a few tenths of a point of the pinned reading.

Each counterfactual with the design pinned and with it chosen, at three values of the IES. Acute ageing moves the chosen design toward earnings-related benefits and France's voting patterns toward flat ones at every IES, the voting patterns furthest at $\rho = 2$, where choosing the design also adds to the rise in the tax rate and shortens the workweek; the direction in which France's income distribution moves the design changes with the IES, and the last row follows the voting patterns.

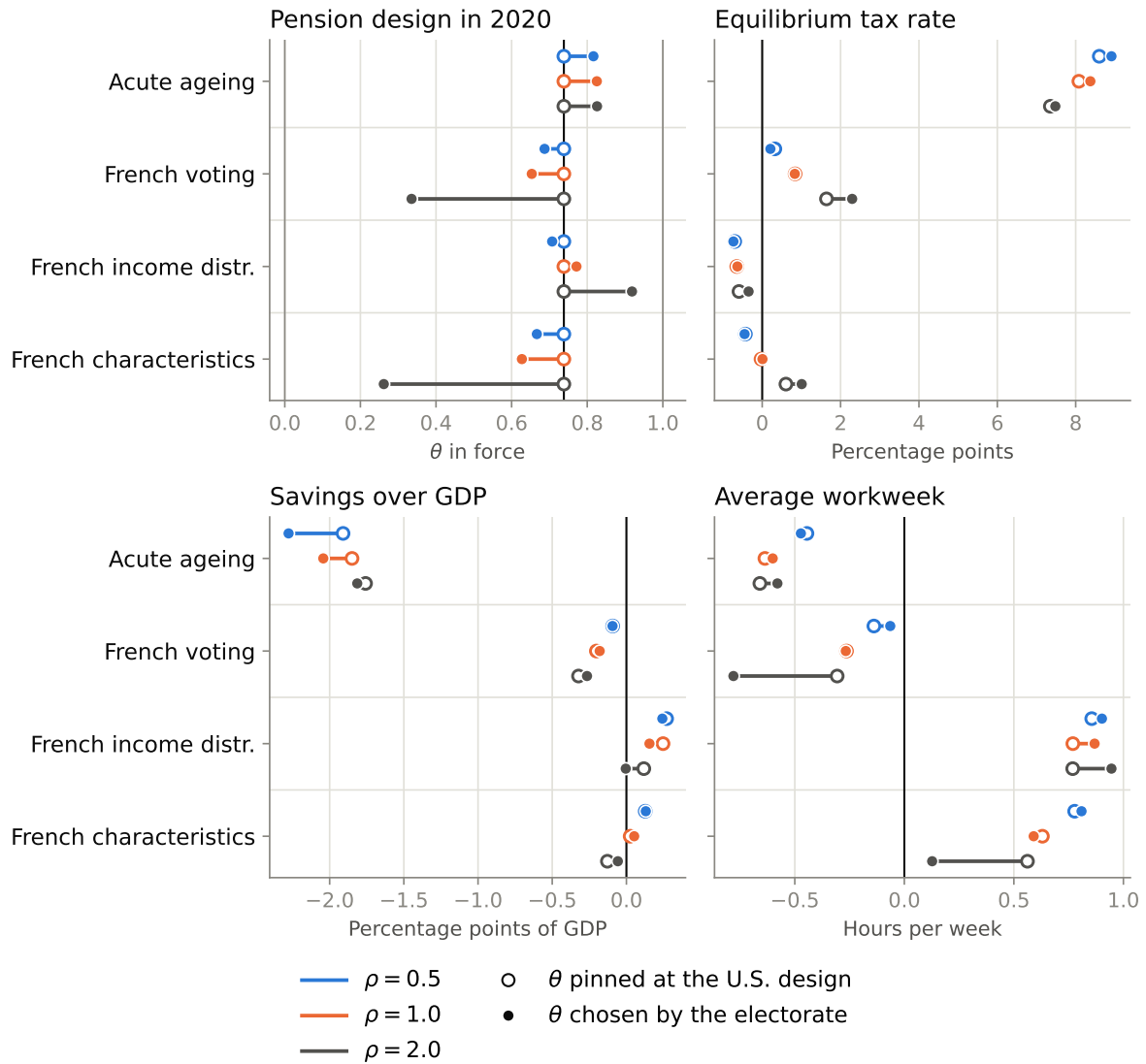
Table OA.42: Endogenous pension design across counterfactuals and intertemporal elasticities in the U.S.

Scenario	$\rho = 0.5$			$\rho = 1.0$			$\rho = 2.0$		
	θ chosen	Tax, pinned	Tax, chosen	θ chosen	Tax, pinned	Tax, chosen	θ chosen	Tax, pinned	Tax, chosen
Baseline	0.738	14.43%	14.41%	0.738	14.43%	14.43%	0.738	14.43%	14.44%
Acute ageing	0.816	23.01%	23.32%	0.825	22.51%	22.80%	0.826	21.79%	21.92%
Income distribution	0.707	13.71%	13.67%	0.772	13.78%	13.80%	0.918	13.85%	14.09%
Voting patterns	0.687	14.73%	14.62%	0.653	15.27%	15.26%	0.335	16.08%	16.73%
French characteristics	0.667	13.98%	13.96%	0.627	14.40%	14.44%	0.262	15.04%	15.45%

Note: The deadweight cost $f(\theta_t, \tau_t)$ of equation (15) of the paper is quadratic in the implicit tax the flat component levies and scaled by the size of the system, $f = \exp\{-\frac{1}{2}\lambda\tau_t\tilde{V}(1-\theta_t)^2\}$, with λ calibrated per ρ (table OA.41). Every row is a separate equilibrium path read in 2020, with the changed characteristic holding over the whole horizon and the political choice binding from the first period. “Pinned” holds the design at the U.S. value θ^* and “chosen” lets the electorate choose it one period in advance, both under the cost. The composite imposes France's income distribution and voting patterns together. France's level of X , which all French characteristics add, leaves every cell of the row unchanged except the last digit of the tax at $\rho = 0.5$, so up to that digit it is also the design and the tax under all French characteristics. The savings and workweek readings are in figure the paper and the tables of section OA.4.2.

Table OA.43: acute ageing at each IES, read with the design pinned at the U.S. value and with it chosen, both under the cost. The baseline rows are the economy with the cost in place and the design chosen, so they differ slightly from those of section OA.3.4. The chosen design is

Figure OA.11: Endogenous versus exogenous pension design across counterfactuals



Note: The first panel is the design in 2020 as a level, with the muted lines at the corners $\theta = 0$ and $\theta = 1$; the other three are deviations from the endogenous- θ baseline, whose tax rate is the baseline row of table OA.42 and whose savings rate and workweek are the baseline rows of the tables of this section. The savings rate is savings relative to GDP. An open marker is the reading with the design pinned at the U.S. value, a filled one the reading with the design chosen.

more Bismarckian at every IES and nearly the same across them, and choosing it adds a little to the rise in the tax rate and to the fall in the savings rate and leaves the workweek almost unchanged.

Table OA.43: Endogenous design and “acute ageing” in the U.S.

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.738	14.41%	16.10%	39.45
	1.0	0.738	14.43%	15.41%	39.47
	2.0	0.738	14.44%	14.87%	39.42
Exogenous θ	0.5	0.738	23.01%	−1.91 p.p.	39.00
	1.0	0.738	22.51%	−1.85 p.p.	38.83
	2.0	0.738	21.79%	−1.76 p.p.	38.76
Endogenous θ	0.5	0.816	23.32%	−2.28 p.p.	38.97
	1.0	0.825	22.80%	−2.04 p.p.	38.87
	2.0	0.826	21.92%	−1.81 p.p.	38.84

Note: The deadweight cost $f(\theta_t, \tau_t)$ of equation (15) of the paper is quadratic in the implicit tax the flat component levies and scaled by the size of the system, $f = \exp\{-\frac{1}{2}\lambda\tau_t\tilde{V}(1-\theta_t)^2\}$, with λ calibrated per ρ (table OA.41). Each ρ is separately calibrated and its workweek normalised against its own baseline. The savings rate is savings relative to GDP; the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. The “acute ageing” scenario sets $\nu_t = 1$ throughout.

France's income distribution on the U.S. at each IES, read with the design pinned and with it chosen. With the design pinned the tax rate falls at every IES, by less than without the cost (table OA.31), since the compressed distribution also lowers the share of revenue the flat component dissipates. The chosen design is less Bismarckian than the observed one at $\rho = 0.5$ and more Bismarckian at $\rho = 1$ and 2, and choosing it keeps the tax rate close to the pinned reading at $\rho = 0.5$ and 1 and raises it at $\rho = 2$.

France's voting patterns on the U.S. at each IES, read with the design pinned and with it chosen. The chosen design moves toward flat benefits at every IES, furthest at $\rho = 2$, where the cost that holds the observed design in place is thinnest; choosing it keeps the tax rate close to the pinned reading at $\rho = 0.5$ and 1, and at $\rho = 2$ raises it and shortens the workweek.

All of France's household characteristics on the U.S. at each IES, read with the design pinned and with it chosen. The voting patterns prevail: the chosen design moves toward flat benefits at every IES, further than under the voting patterns alone (table OA.45), and choosing it keeps the tax rate close to the pinned reading at $\rho = 0.5$ and 1 and raises it at $\rho = 2$.

Table OA.44: Endogenous design and the French income distribution in the U.S.

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.738	14.41%	16.10%	39.45
	1.0	0.738	14.43%	15.41%	39.47
	2.0	0.738	14.44%	14.87%	39.42
Exogenous θ	0.5	0.738	13.71%	+0.27 p.p.	40.30
	1.0	0.738	13.78%	+0.25 p.p.	40.24
	2.0	0.738	13.85%	+0.12 p.p.	40.19
Endogenous θ	0.5	0.707	13.67%	+0.24 p.p.	40.35
	1.0	0.772	13.80%	+0.16 p.p.	40.34
	2.0	0.918	14.09%	0.00 p.p.	40.36
France	0.5	1.000	21.29%	−1.44 p.p.	35.24
	1.0	1.000	21.29%	−1.97 p.p.	35.24
	2.0	1.000	21.29%	−2.39 p.p.	35.24

Note: The cost specification and the units are those of table OA.43. The France row is France's own calibrated path rather than a counterfactual on the U.S. model: it carries France's own ω as well as its characteristics, and its workweek is a calibration target rather than a prediction.

Table OA.45: Endogenous design and French voting patterns in the U.S.

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.738	14.41%	16.10%	39.45
	1.0	0.738	14.43%	15.41%	39.47
	2.0	0.738	14.44%	14.87%	39.42
Exogenous θ	0.5	0.738	14.73%	−0.09 p.p.	39.31
	1.0	0.738	15.27%	−0.20 p.p.	39.21
	2.0	0.738	16.08%	−0.32 p.p.	39.11
Endogenous θ	0.5	0.687	14.62%	−0.09 p.p.	39.38
	1.0	0.653	15.26%	−0.18 p.p.	39.20
	2.0	0.335	16.73%	−0.27 p.p.	38.64
France	0.5	1.000	21.29%	−1.44 p.p.	35.24
	1.0	1.000	21.29%	−1.97 p.p.	35.24
	2.0	1.000	21.29%	−2.39 p.p.	35.24

Note: The cost specification and the units are those of table OA.43. The France row is France's own calibrated path rather than a counterfactual on the U.S. model: it carries France's own ω as well as its characteristics, and its workweek is a calibration target rather than a prediction.

Table OA.46: Endogenous design and all French characteristics in the U.S.

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.738	14.41%	16.10%	39.45
	1.0	0.738	14.43%	15.41%	39.47
	2.0	0.738	14.44%	14.87%	39.42
Exogenous θ	0.5	0.738	13.97%	+0.13 p.p.	34.66
	1.0	0.738	14.40%	+0.03 p.p.	34.96
	2.0	0.738	15.04%	−0.13 p.p.	35.22
Endogenous θ	0.5	0.667	13.95%	+0.13 p.p.	34.69
	1.0	0.627	14.44%	+0.05 p.p.	34.93
	2.0	0.262	15.45%	−0.06 p.p.	34.83
France	0.5	1.000	21.29%	−1.44 p.p.	35.24
	1.0	1.000	21.29%	−1.97 p.p.	35.24
	2.0	1.000	21.29%	−2.39 p.p.	35.24

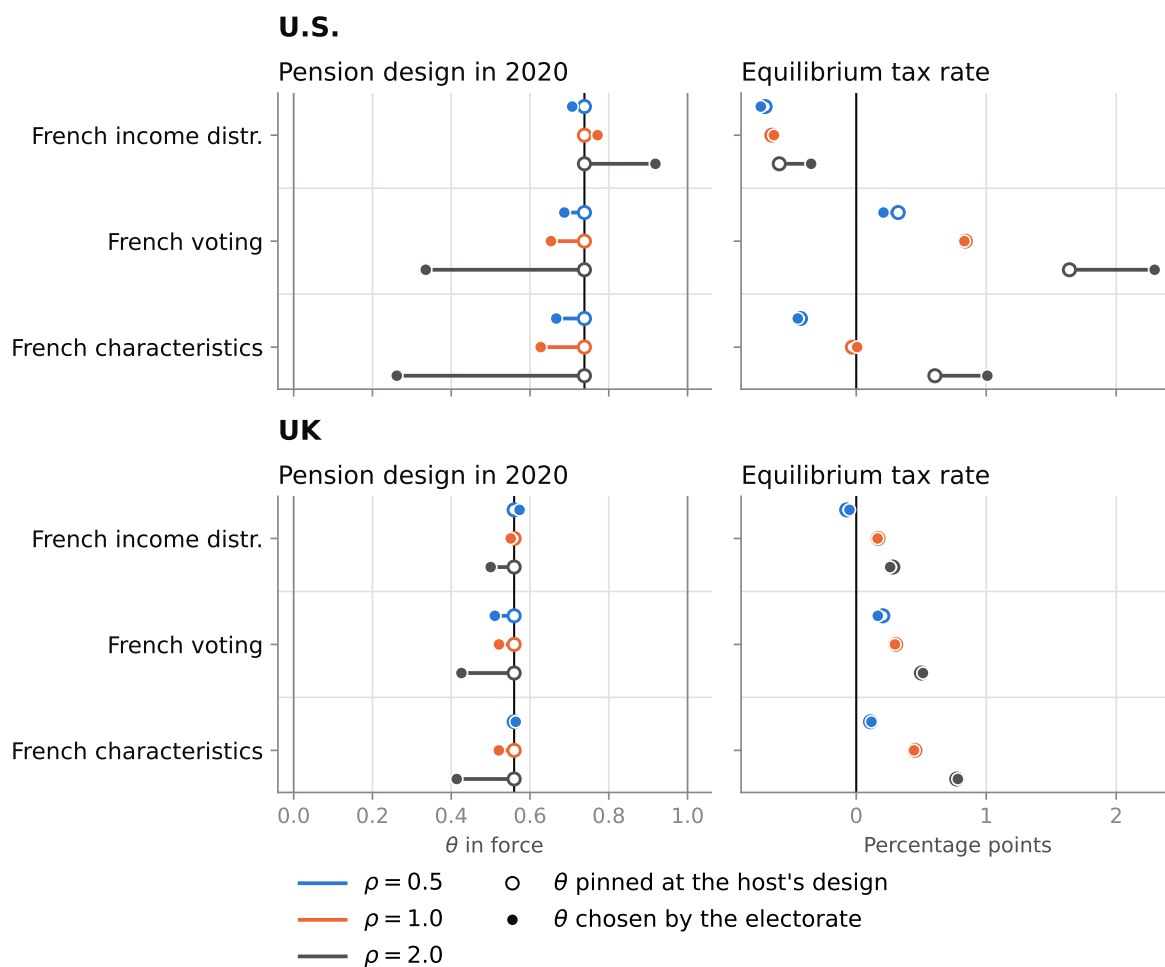
Note: The cost specification and the units are those of table OA.43. The scenario replaces the U.S. η_i , the level of X_i and the voting weights μ_i with France's simultaneously. The France row is France's own calibrated path rather than a counterfactual on the U.S. model: it carries France's own ω as well as its characteristics, and its workweek is a calibration target rather than a prediction.

OA.4.3 The UK as host

The same counterfactuals with the UK as host, at the UK's own cost parameter, the one at which its electorate re-elects the UK's observed design; it lies below the U.S. parameter at an IES of one or below and above it at two. The figure sets the two hosts side by side; the tables print the UK's numbers. On both hosts France's voting profile moves the design most, toward flat benefits, and by less on the UK, whose own voting profile is already flatter than the U.S. one. France's income distribution moves the two hosts' designs in opposite directions, since on the U.S. it compresses relative incomes and on the UK it spreads them. The design response leaves the UK's tax rate, savings rate and workweek essentially at the reading with the design pinned, so on the UK the response is confined to the design itself.

France's characteristics on the two hosts with the design chosen, the U.S. in the upper row of panels and the UK in the lower, on shared axes so that the hosts compare by position. The last row is France's income distribution and voting patterns together, whose design and tax rate are those of the tables' all-French rows. On the U.S. that row follows the voting patterns at every IES; on the UK it does so at $\rho = 1$ and 2, while at $\rho = 0.5$ the chosen design barely moves.

Figure OA.12: Endogenous pension design under French characteristics in the U.S. and the UK



Note: Each host at its own cost parameter. The first column is the design in force in 2020 as a level, the reference line at the host's observed design and the muted lines at the corners; the second is the tax rate as a deviation from the host's endogenous- θ baseline, whose levels are the baseline rows of the tables of this section. An open marker is the reading with the design pinned at the host's value, a filled one the reading with the design chosen.

France's income distribution on the UK at each IES, read with the design pinned at the UK's value and with it chosen. The chosen design moves toward earnings-related benefits at $\rho = 0.5$ and toward flat ones at $\rho = 1$ and 2, each time the reverse of its move on the U.S. (table OA.44).

Table OA.47: Endogenous design and the French income distribution in the UK

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.560	11.86%	19.11%	34.73
	1.0	0.560	11.86%	17.25%	34.75
	2.0	0.560	11.86%	15.81%	34.74
Exogenous θ	0.5	0.560	11.78%	+0.11 p.p.	34.63
	1.0	0.560	12.03%	−0.03 p.p.	34.58
	2.0	0.560	12.14%	−0.06 p.p.	34.56
Endogenous θ	0.5	0.574	11.81%	+0.03 p.p.	34.63
	1.0	0.551	12.02%	−0.04 p.p.	34.59
	2.0	0.500	12.12%	−0.03 p.p.	34.53
France	0.5	1.000	21.29%	−4.45 p.p.	35.24
	1.0	1.000	21.29%	−3.81 p.p.	35.24
	2.0	1.000	21.29%	−3.34 p.p.	35.24

Note: The cost specification and the units are those of table OA.43, at the UK's own cost parameter, calibrated per ρ so that the UK electorate re-elects its observed design $\theta^* = 0.560$: $\lambda = 15.117, 7.264, 2.777$ at $\rho = 0.5, 1.0, 2.0$, with β imposed from the U.S. and ω recalibrated at each trial value. France's income groups are cut at the UK's own income percentiles (table 9 of the paper). The France row is France's own calibrated path rather than a counterfactual on the UK model: it carries France's own ω as well as its characteristics, and its workweek is a calibration target rather than a prediction.

France's voting patterns on the UK at each IES, read with the design pinned at the UK's value and with it chosen. The chosen design moves toward flat benefits at every IES, furthest at $\rho = 2$, as on the U.S. (table OA.45).

All of France's household characteristics on the UK at each IES, read with the design pinned at the UK's value and with it chosen. At $\rho = 1$ and 2 the chosen design follows the voting patterns toward flat benefits, while at $\rho = 0.5$ it barely moves; the workweek ends above the UK's baseline in both readings, where on the U.S. it ends below.

Table OA.48: Endogenous design and French voting patterns in the UK

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.560	11.86%	19.11%	34.73
	1.0	0.560	11.86%	17.25%	34.75
	2.0	0.560	11.86%	15.81%	34.74
Exogenous θ	0.5	0.560	12.06%	−0.08 p.p.	34.68
	1.0	0.560	12.16%	−0.09 p.p.	34.67
	2.0	0.560	12.36%	−0.11 p.p.	34.65
Endogenous θ	0.5	0.511	12.02%	−0.08 p.p.	34.69
	1.0	0.521	12.15%	−0.08 p.p.	34.66
	2.0	0.426	12.37%	−0.09 p.p.	34.60
France	0.5	1.000	21.29%	−4.45 p.p.	35.24
	1.0	1.000	21.29%	−3.81 p.p.	35.24
	2.0	1.000	21.29%	−3.34 p.p.	35.24

Note: The cost specification, the cost parameter and the units are those of table OA.47. The France row is France's own calibrated path rather than a counterfactual on the UK model: it carries France's own ω as well as its characteristics, and its workweek is a calibration target rather than a prediction.

Table OA.49: Endogenous design and all French characteristics in the UK

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.560	11.86%	19.11%	34.73
	1.0	0.560	11.86%	17.25%	34.75
	2.0	0.560	11.86%	15.81%	34.74
Exogenous θ	0.5	0.560	11.97%	0.00 p.p.	35.75
	1.0	0.560	12.31%	−0.14 p.p.	35.66
	2.0	0.560	12.63%	−0.17 p.p.	35.63
Endogenous θ	0.5	0.564	11.98%	−0.05 p.p.	35.76
	1.0	0.521	12.30%	−0.13 p.p.	35.65
	2.0	0.414	12.64%	−0.12 p.p.	35.53
France	0.5	1.000	21.29%	−4.45 p.p.	35.24
	1.0	1.000	21.29%	−3.81 p.p.	35.24
	2.0	1.000	21.29%	−3.34 p.p.	35.24

Note: The cost specification, the cost parameter and the units are those of table OA.47. The scenario replaces the UK η_i , the level of X_i and the voting weights μ_i with France's simultaneously. The France row is France's own calibrated path rather than a counterfactual on the UK model: it carries France's own ω as well as its characteristics, and its workweek is a calibration target rather than a prediction.

OA.4.4 The design along the baseline path

The calibration pins the design in 2020 only; every other design on the path, before 2020 as after it, is an outcome of the model. The paper's figure is the projection from 2020 at an IES of one; here are the full paths of the design and of the tax rate from the first period, chosen and pinned, at each IES and at two other values of the Frisch elasticity ξ , which the paper reports in a footnote.

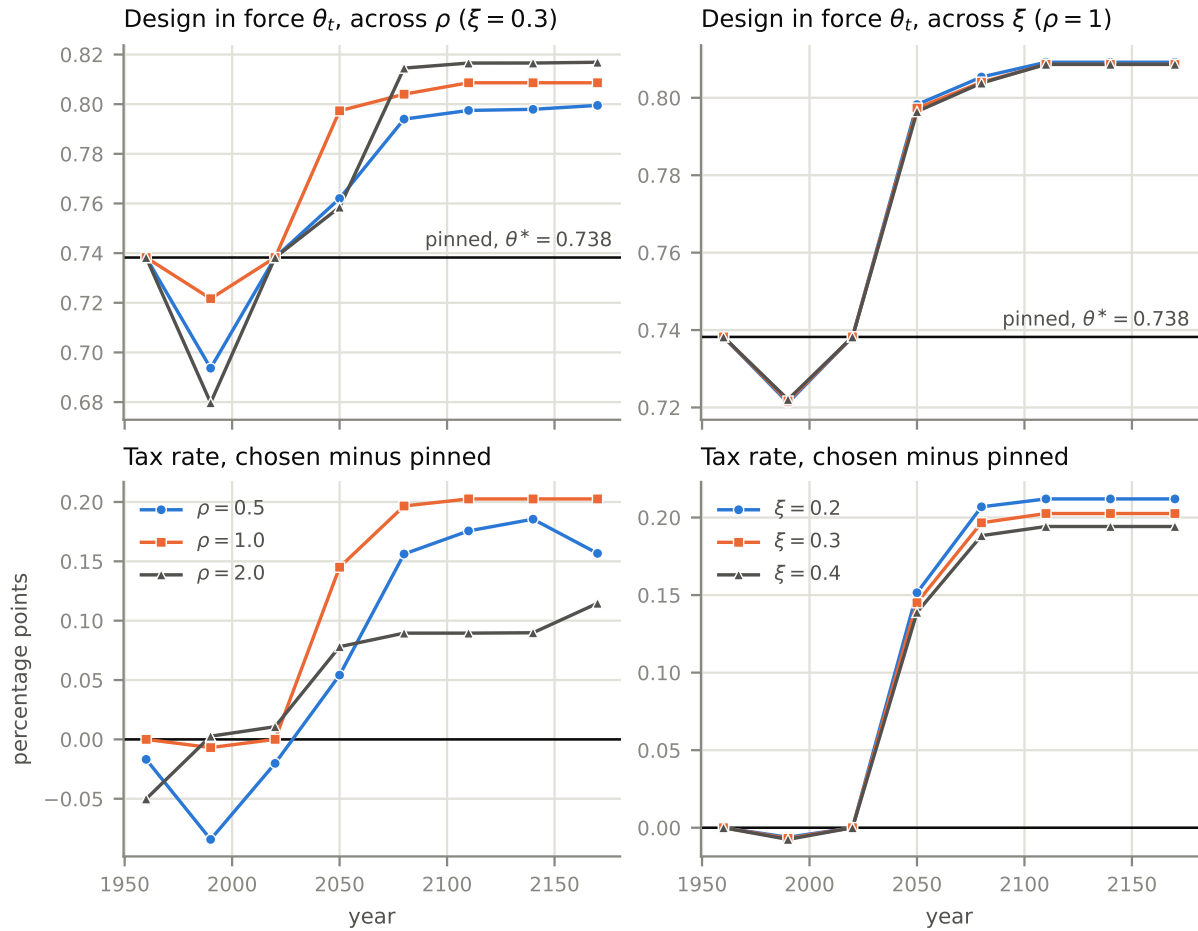
Beside these, the site shows the paper's figure 7.1 (section 7), which this edition does not reprint.

Figure 7.1 of the paper: the paper's figure, the projection from the calibration year at an IES of one. Workers per retiree fall until 2080 and are held constant after; the chosen design rises with the fall and levels off with it, and the tax rate with the design chosen lies a little above the one with it pinned.

Along the baseline path the chosen design dips below the observed one before 2020 at every specification, returns to it in 2020, where the calibration pins it, and then rises toward earnings-related benefits as ageing proceeds, leveling off with population growth. After 2020 the tax rate with the design chosen lies slightly above the one with it pinned, since a more Bismarckian system dissipates less and is worth more to the electorate at the margin, and the Frisch elasticity barely moves either line.

The design choice at three Frisch elasticities and an IES of one, each recalibrated with its own cost parameter; the design channel runs through hours, which makes this the elasticity to vary. The cost parameter falls a little as the elasticity rises, the designs on the baseline path and under acute ageing are nearly unchanged, and only the level of the tax rate under acute ageing moves, lower at a higher elasticity whether the design is pinned or chosen.

Figure OA.13: The design and the tax along the baseline path, U.S.



Note: The design chosen one period in advance from the first period on, and the tax rate with the design chosen and with it pinned at θ^* , along the baseline path at each IES and, at $\rho = 1$, at the Frisch elasticities $\xi = 0.2$ and 0.4; the tables of this section print the values.

Table OA.50: The design and the tax along the baseline path, chosen and pinned

Year	θ_t chosen	Tax rate, chosen	Tax rate, pinned	Difference
$\rho = 0.5$				
1960	0.738	13.28%	13.29%	−0.02 p.p.
1990	0.694	13.16%	13.25%	−0.08 p.p.
2020	0.738	14.41%	14.43%	−0.02 p.p.
2050	0.762	19.67%	19.62%	+0.05 p.p.
2080	0.794	21.24%	21.09%	+0.16 p.p.
2110	0.797	21.29%	21.11%	+0.18 p.p.
2140	0.798	21.33%	21.14%	+0.19 p.p.
2170	0.800	21.51%	21.35%	+0.16 p.p.
$\rho = 1.0$				
1960	0.738	13.11%	13.11%	0.00 p.p.
1990	0.722	13.10%	13.11%	−0.01 p.p.
2020	0.738	14.43%	14.43%	0.00 p.p.
2050	0.797	19.48%	19.33%	+0.15 p.p.
2080	0.804	20.87%	20.67%	+0.20 p.p.
2110	0.809	20.88%	20.67%	+0.20 p.p.
2140	0.809	20.88%	20.67%	+0.20 p.p.
2170	0.809	20.88%	20.67%	+0.20 p.p.
$\rho = 2.0$				
1960	0.738	13.02%	13.07%	−0.05 p.p.
1990	0.680	13.08%	13.08%	0.00 p.p.
2020	0.738	14.44%	14.43%	+0.01 p.p.
2050	0.758	19.04%	18.96%	+0.08 p.p.
2080	0.815	20.23%	20.14%	+0.09 p.p.
2110	0.817	20.23%	20.14%	+0.09 p.p.
2140	0.817	20.23%	20.14%	+0.09 p.p.
2170	0.817	20.31%	20.19%	+0.11 p.p.
$\rho = 1.0, \xi = 0.2$				
1960	0.738	13.05%	13.05%	0.00 p.p.
1990	0.721	13.04%	13.05%	−0.01 p.p.
2020	0.738	14.43%	14.43%	0.00 p.p.
2050	0.798	19.73%	19.58%	+0.15 p.p.
2080	0.805	21.20%	20.99%	+0.21 p.p.
2110	0.809	21.20%	20.99%	+0.21 p.p.
2140	0.809	21.20%	20.99%	+0.21 p.p.
2170	0.809	21.20%	20.99%	+0.21 p.p.
$\rho = 1.0, \xi = 0.4$				
1960	0.738	13.16%	13.16%	0.00 p.p.
1990	0.722	13.16%	13.16%	−0.01 p.p.
2020	0.738	14.43%	14.43%	0.00 p.p.
2050	0.796	19.24%	19.19%	+0.14 p.p.

Table OA.51: The endogenous design across the Frisch elasticity ξ

ξ	λ	Design on the baseline path			Acute ageing, 2020		
		θ_{2050}	θ_{2080}	θ_{2110}	θ chosen	Tax, pinned	Tax, chosen
0.2	8.736	0.798	0.805	0.809	0.825	22.94%	23.24%
0.3	8.643	0.797	0.804	0.809	0.825	22.51%	22.80%
0.4	8.547	0.796	0.804	0.809	0.826	22.11%	22.39%

Note: Log preferences ($\rho = 1$). At each Frisch elasticity ξ the model is recalibrated, (β, ω, X) to their targets and λ so that the design in force in 2020 is $\theta^* = 0.738$. The design columns follow the baseline path of table OA.50. Acute ageing sets $\nu_t = 1$ throughout, a new equilibrium path read at 2020, with the design pinned at θ^* or chosen; at $\xi = 0.3$ it is the row of table OA.43 at $\rho = 1$.

OA.4.5 The timing of the design choice

Appendix C.1 of the paper shows that each timing of the choice puts the design at a corner when redistribution is costless: a design chosen together with the tax rate re-splits benefits current retirees have already earned, a design chosen one period ahead trades the young's redistributive stake against the forward-looking one, and a design chosen once and for all enters every channel at once. The table reports the numerical checks of these timings.

One block per timing of the choice, all without the cost of redistribution. Chosen with the tax or one period in advance, the design is the Beveridgean corner at every IES checked. Chosen once and for all, it is the Beveridgean corner at the lower elasticities of the grid shown and the Bismarckian corner at the higher ones; near the switch the electorate is close to indifferent between the two, so the switch is not sharply located.

Table OA.52: The timing of the design choice without a cost

Timing	CRRA (ρ)	θ chosen	Tax rate	Largest $\partial \mathcal{W}_t / \partial \theta_t$
With the tax	0.5	0.000	–	–1.15
	2.0	0.000	–	–0.00125
One period in advance	1.0	0.000	–	–
Once and for all	1.0	0.000	18.52%	–
	1.1	0.000	18.16%	–
	1.2	0.000	17.82%	–
	1.3	0.000	17.47%	–
	1.4	1.000	12.90%	–
	1.5	1.000	13.04%	–
	2.0	1.000	13.72%	–

Note: No cost of redistribution; $\rho = 1$ is log preferences. With the tax: the largest value of the marginal effect of the design on the political objective, equation (C.1) of the paper, over a grid of θ_t on $[0, 1]$ and the dates 1990–2170 of the baseline path; it is negative everywhere, so the choice is the corner $\theta_t = 0$ at every date. Its scale is that of the objective, which differs across ρ ; the log case is checked in the regression suite of the code. One period in advance: the design in force in 2020 on the path where the choice binds from the first period. Once and for all: the design chosen in 2020, with the savings of the period before made in anticipation of the vote, and the tax rate at that design. Pinning the savings ratio at the incumbent design instead, an unanticipated vote, gives the same design in every row.

OA.4.6 An alternative cost specification

A cost attached to the design rather than to the transfer it performs. The table sets it beside the paper's cost, which scales with the dispersion of relative incomes: the calibration of each, and the design each lets the electorate choose under acute ageing and under France's income distribution and voting patterns.

The paper's cost in the upper block and the cost on the design in the lower. Under both, acute ageing makes the chosen design more Bismarckian and France's voting patterns less so at every IES, the first by less and the second by more under the cost on the design. Under the cost on

the design, France's income distribution, which compresses the redistributive stakes but leaves that cost in place, takes the design to the Bismarckian corner at every IES.

Table OA.53: Two specifications of the deadweight cost: calibration and the chosen design

Cost on	CRRA (ρ)	Calibration				θ chosen (2020)		
		λ, p	θ^*	$f(\theta^*)$	$f(0)$	Acute age- ing	French in- come dis- tribu- tion	French vot- ing pat- terns
redistribution	0.5	18.267	0.738	0.961	0.563	0.816	0.707	0.687
	1.0	8.643	0.738	0.982	0.762	0.825	0.772	0.653
	2.0	1.724	0.738	0.996	0.947	0.826	0.918	0.335
the design	0.5	0.935	0.738	0.876	0.500	0.756	1.000	0.654
	1.0	0.408	0.738	0.942	0.500	0.778	1.000	0.533
	2.0	0.085	0.738	0.987	0.500	0.821	1.000	0.259

Note: Two specifications of the deadweight cost f of equation (15) of the paper, each with one parameter calibrated per ρ so that the design in force in 2020 is the observed θ^* , with (β, ω) recalibrated at each trial value. The cost on redistribution is that of table OA.41, $f(\theta, \tau) = \exp\{-\frac{1}{2}\lambda\tau\tilde{V}(1-\theta)^2\}$: quadratic in the implicit tax the flat component levies and scaled by the size of the system, so nothing is lost when nothing is redistributed. The cost on the design is $f(\theta) = \phi + (1-\phi)\theta^p$ with $\phi = 0.5$ imposed: a flat benefit forfeits a share $1-\phi$ of revenue whatever it redistributes. $f(\theta^*)$ and $f(0)$ are the shares of revenue that reach households at the observed design and at a flat benefit; the cost on redistribution is read at the 2020 tax rate. The last three columns are the design in force in 2020 when the electorate chooses it; under the cost on redistribution they are the endogenous- θ rows of tables OA.43, OA.44 and OA.45.

OA.5 Numerical checks

Section 4 of the paper computes date-specific Markov policy functions by backward induction along the demographic transition, and selects among the candidate equilibria at each state; the technical documentation (Berg and Gonzalez-Eiras 2026) gives the algorithms and the code. This chapter reports two checks of that computation.

OA.5.1 Date-specific and stationary policies

The table prints the gap between the date-specific solution and a time-invariant policy function, the approximation of much of the quantitative politico-economic literature, at the same dates. Section 4 summarizes it in a footnote.

One row per policy and IES, one column per date of the path. Argentina's informal savers add a state, and a design chosen at t is weighed against the electorate of $t + 1$, so both differ from the stationary approximation even at an IES of one. The tax gaps are largest at the lowest IES, in Argentina above all, and every gap shrinks toward zero late in the horizon, apart from the last period shown.

Table OA.54: Stationary against date-specific policy functions

Policy	ρ	Stationary minus date-specific								
<i>United States</i>		1960	1990	2020	2050	2080	2110	2140	2170	
Tax rate, design given (p.p.)	0.5	+0.01	+0.04	+0.17	+0.01	−0.03	−0.01	−0.02	−0.26	
	0.7	0.00	+0.02	+0.08	0.00	−0.02	−0.01	−0.01	−0.07	
	1.3	0.00	−0.01	−0.04	+0.01	+0.02	0.00	0.00	0.00	
	1.5	0.00	−0.01	−0.06	+0.02	+0.02	+0.01	0.00	−0.01	
	2.0	0.00	−0.01	−0.07	+0.03	+0.03	+0.01	0.00	−0.06	
Tax rate, design chosen (p.p.)	0.5	+0.01	+0.06	+0.13	−0.02	−0.08	−0.01	−0.03	−0.19	
	1.0	0.00	0.00	−0.02	−0.12	−0.01	0.00	0.00	0.00	
	2.0	0.00	−0.05	−0.10	−0.03	+0.02	+0.01	0.00	−0.08	
Design chosen (θ)	0.5	0.000	+0.003	−0.042	−0.022	−0.022	0.000	0.000	−0.002	
	1.0	0.000	0.000	−0.017	−0.053	−0.004	0.000	0.000	0.000	
	2.0	0.000	−0.001	−0.059	−0.020	−0.041	0.000	0.000	0.000	
<i>Argentina</i>		1920	1950	1980	2010	2040	2070	2100	2130	2160
Tax rate (p.p.)	0.5	+0.15	+0.49	+0.48	−0.20	+0.34	+0.16	−0.08	−0.04	−0.63
	1.0	−0.01	0.00	−0.08	−0.06	+0.16	+0.19	−0.02	−0.01	−0.20
	2.0	+0.01	+0.01	−0.11	−0.16	+0.04	+0.21	0.00	0.00	−0.15

Note: Each cell is the policy on the path walked by time-invariant policy functions, each solved at its own date's ν_t and applied from the same initial state, minus the date-specific policy of the paper's solution: the tax rate in percentage points, the chosen design in units of θ . Under log preferences ($\rho = 1$) with the design given the two coincide and the U.S. block has no row. The design chosen is at the paper's cost specification. The terminal period of the finite horizon is not shown; the last period shown carries the date-specific solution's approach to it (appendix D of the paper) rather than the demographic transition.

OA.5.2 Equilibrium selection

The table prints, for every file of results behind an exhibit, the largest number of equilibria and of candidate solutions found at any state, and the number of states at which no candidate passed the equilibrium test and the solver fell back on the integral rule of the technical documentation. A single equilibrium at every state means that the selection rule of section 4 never had to choose. A file prints no count where its solve selects no tax, the economic-equilibrium paths, or where its driver did not record the count: the calibration grids of the U.S., the UK and France, the Frisch-elasticity and timing files, and the tax layer of the CRRA endogenous-design calibration and path files, whose design layer is counted in the lower panel. The paper's statement in section 4 rests on the files that record the count.

Table OA.55: Equilibrium counts in the results the exhibits read

File	Rows	Counted	Equilibria, max	Candidates, max	Fallbacks
<i>Tax rule: Argentina</i>					
informalSavings_rhoGridCommonX.csv	16	16	1	3	0
universal_match_rho*_commonX.csv	112	112	1	3	0
(16 files)					
eeOnly_match_rho*_commonX.csv	112	0	—	—	—
(16 files)					
epsThetaGrid_rho1.0000_commonX.csv	378	378	1	3	0
informalSavings_rhoGrid.csv	16	16	1	3	0
universal_match_rho*.csv (16 files)	112	112	1	3	0
eeOnly_match_rho*.csv (16 files)	112	0	—	—	—
epsThetaGrid_rho1.0000.csv	378	378	1	3	0
ARG_stationaryApprox_commonX.csv	30	30	1	3	0
<i>Tax rule: the U.S., the UK and France</i>					
US_rhoGridCommonX.csv	16	0	—	—	—
UK_rhoGridCommonX.csv	16	0	—	—	—
FR_rhoGridCommonX.csv	16	0	—	—	—
US_shocksCommonX.csv	54	6	1	3	0
UK_shocksCommonX.csv	54	6	1	3	0
US_rhoGrid.csv	16	0	—	—	—
UK_rhoGrid.csv	16	0	—	—	—
FR_rhoGrid.csv	16	0	—	—	—
US_shocks.csv	54	6	1	3	0
UK_shocks.csv	54	6	1	3	0
escCalibration.csv	5	3	1	3	0
escCalibrationCRRA.csv	8	0	—	—	—
escCalibrationCRRAUK.csv	2	0	—	—	—
escCountry.csv	12	9	1	3	0
escShocks.csv	34	34	1	3	0
escShocksCRRA.csv	120	16	1	3	0
escShocksUK.csv	13	13	1	3	0
escShocksCRRAUK.csv	26	14	1	3	0
escPath.csv	16	16	1	3	0
escPathCRRA.csv	48	0	—	—	—
escXiRobustness.csv	33	0	—	—	—
escPermanent.csv	7	0	—	—	—
escPermanentCRRA.csv	6	0	—	—	—
escSequentialCRRA.csv	14	0	—	—	—
US_stationaryApprox.csv	45	45	1	3	0
US_ESC_stationaryApprox.csv	27	9	1	3	0

OA.6 Replication

Every table and figure of the paper’s quantitative sections and of this appendix, except the paper’s figure 5.1 of Argentina’s beneficiaries, is produced by the pipeline in `python/paper` of the [repository](#), in three stages: the calibrations, the experiments, and a build that reads only their results and writes the exhibits in seconds. `REPLICATION.md` in the repository is the map from data and code to every output, with the commands and the time each stage takes; the technical documentation (Berg and Gonzalez-Eiras 2026) describes the model, its equilibria and the algorithms.

The table lists every exhibit shown here, where it appears and the builder that makes it.

Output	Where	Builder
RobustnessMap	in the paper, appendix G	<code>figuresUS.robustnessMap</code>
OECDdata	in the paper, section 1	<code>oecdFigure1.figure</code>
OECD_Countries	Table OA.1, OA.1.1	<code>tablesOA.oecdCountries</code>
OECD_Sources	Table OA.2, OA.1.1	<code>tablesOA.oecdSources</code>
OECD_Correlations	Table OA.3, OA.1.2	<code>tablesOA.oecdCorrelations</code>
ArgentinaCalibration	in the paper, section 5	<code>tables.argentinaCalibration</code>
ArgentinaCalibration_vectorX	Table OA.4, OA.2.1	<code>tables.argentinaCalibration(commonX = False)</code>
ArgentinaUniversal	in the paper, section 5	<code>tables.argentinaUniversal</code>
ArgentinaUniversal_vectorX	Table OA.5, OA.2.2	<code>tables.argentinaUniversal(commonX = False)</code>
ArgentinaReformPath	Table OA.6, OA.2.2	<code>tablesOA.argentinaReformPath</code>
ArgentinaReformPath_vectorX	Table OA.7, OA.2.2	<code>tablesOA.argentinaReformPath(commonX = False)</code>
ARG_CRRALOG	in the paper, section 5	<code>figures.argCrralLog</code>
ARG_CRRALOG_vectorX	Figure OA.1, OA.2.3	<code>figures.argCrralLog(commonX = False)</code>
ArgentinaReformByRho	Table OA.8, OA.2.3	<code>tables.argentinaReformByRho</code>
ArgentinaReformByRho_vectorX	Table OA.9, OA.2.3	<code>tables.argentinaReformByRho(commonX = False)</code>
ARG_LOG_FourInOne	Figure OA.2, OA.2.4	<code>figures.argLogFourInOne</code>
ARG_LOG_FourInOne_vectorX	Figure OA.3, OA.2.4	<code>figures.argLogFourInOne(commonX = False)</code>
ARG_RhoGrid	Figure OA.4, OA.2.5	<code>figuresOA.argRhoGrid</code>
ARG_RhoGrid_vectorX	Figure OA.5, OA.2.5	<code>figuresOA.argRhoGrid(commonX = False)</code>
ARG_RhoGridTable	Table OA.10, OA.2.5	<code>tablesOA.argRhoGridTable</code>
ARG_RhoGridTable_vectorX	Table OA.11, OA.2.5	<code>tablesOA.argRhoGridTable(commonX = False)</code>
USUKFRCalibration	in the paper, section 6	<code>tablesUS.usukfrCalibration</code>
USUKFRCalibration_vectorX	Table OA.12, OA.3.1	<code>tablesUS.usukfrCalibration(commonX = False)</code>
US_householdheterogeneity	in the paper, appendix F	<code>tablesUS.usHouseholdHeterogeneity</code>
US_householdheterogeneity_vectorX	Table OA.13, OA.3.1	<code>tablesUS.usHouseholdHeterogeneity(commonX = False)</code>
FR_householdheterogeneity	in the paper, appendix F	<code>tablesUS.frHouseholdHeterogeneity</code>
FR_householdheterogeneity_vectorX	Table OA.14, OA.3.1	<code>tablesUS.frHouseholdHeterogeneity(commonX = False)</code>
UK_householdheterogeneity	in the paper, appendix F	<code>tablesUS.ukHouseholdHeterogeneity</code>
UK_householdheterogeneity_vectorX	Table OA.15, OA.3.1	<code>tablesUS.ukHouseholdHeterogeneity(commonX = False)</code>

Output	Where	Builder
FRUK_householdheterogeneity	in the paper, appendix F	tablesUS.frukHouseholdHeterogeneity
FRUK_householdheterogeneity_vectorX	Table OA.16, OA.3.1	tablesUS.frukHouseholdHeterogeneity(commonX = False)
UKUS_householdheterogeneity	Table OA.17, OA.3.1	tablesUS.ukusHouseholdHeterogeneity
UKUS_householdheterogeneity_vectorX	Table OA.18, OA.3.1	tablesUS.ukusHouseholdHeterogeneity(commonX = False)
OECD_RhoGrid	Figure OA.6, OA.3.3	figuresOA.oecdRhoGrid
OECD_RhoGrid_vectorX	Figure OA.7, OA.3.3	figuresOA.oecdRhoGrid(commonX = False)
OECD_RhoGridTable	Table OA.19, OA.3.3	tablesOA.oecdRhoGridTable
OECD_RhoGridTable_vectorX	Table OA.20, OA.3.3	tablesOA.oecdRhoGridTable(commonX = False)
US_overview	in the paper, section 6	figuresUS.usOverview
US_overview_vectorX	Figure OA.8, OA.3.4	figuresUS.usOverview(commonX = False)
US_PensChars	Table OA.21, OA.3.4	tablesUS.usPensChars
US_PensChars_vectorX	Table OA.22, OA.3.4	tablesUS.usPensChars(commonX = False)
US_Ageing	Table OA.23, OA.3.4	tablesUS.usAgeing
US_Ageing_vectorX	Table OA.24, OA.3.4	tablesUS.usAgeing(commonX = False)
US_OtherShocks	Table OA.25, OA.3.4	tablesUS.usOtherShocks
US_OtherShocks_vectorX	Table OA.26, OA.3.4	tablesUS.usOtherShocks(commonX = False)
US_CRRA_PensChars	Table OA.27, OA.3.4	tablesUS.usCrRaPensChars
US_CRRA_PensChars_vectorX	Table OA.28, OA.3.4	tablesUS.usCrRaPensChars(commonX = False)
US_CRRA_Ageing	Table OA.29, OA.3.4	tablesUS.usCrRaAgeing
US_CRRA_Ageing_vectorX	Table OA.30, OA.3.4	tablesUS.usCrRaAgeing(commonX = False)
US_CRRA_OtherShocks	Table OA.31, OA.3.4	tablesUS.usCrRaOtherShocks
US_CRRA_OtherShocks_vectorX	Table OA.32, OA.3.4	tablesUS.usCrRaOtherShocks(commonX = False)
UKUS_French	Figure OA.9, OA.3.5	figuresUS.ukusFrench
UKUS_French_vectorX	Figure OA.10, OA.3.5	figuresUS.ukusFrench(commonX = False)
UK_OtherShocks	Table OA.33, OA.3.5	tablesUS.ukOtherShocks
UK_OtherShocks_vectorX	Table OA.34, OA.3.5	tablesUS.ukOtherShocks(commonX = False)
UK_CRRA_OtherShocks	Table OA.35, OA.3.5	tablesUS.ukCrRaOtherShocks
UK_CRRA_OtherShocks_vectorX	Table OA.36, OA.3.5	tablesUS.ukCrRaOtherShocks(commonX = False)
UK_CRRA_PensChars	Table OA.37, OA.3.5	tablesUS.ukCrRaPensChars
UK_CRRA_PensChars_vectorX	Table OA.38, OA.3.5	tablesUS.ukCrRaPensChars(commonX = False)
UK_CRRA_Ageing	Table OA.39, OA.3.5	tablesUS.ukCrRaAgeing
UK_CRRA_Ageing_vectorX	Table OA.40, OA.3.5	tablesUS.ukCrRaAgeing(commonX = False)
US_ESC_Calibration	Table OA.41, OA.4.1	tablesUS.escCalibrationTable
US_ESC_Country	in the paper, section 7	tablesUS.escCountryTable
US_ESC_Results	in the paper, section 7	tablesUS.escResults
US_ESC_overview	Figure OA.11, OA.4.2	figuresUS.escOverview
US_ESC_Summary	Table OA.42, OA.4.2	tablesUS.escSummary
US_ESC_Ageing	Table OA.43, OA.4.2	tablesUS.escAgeing
US_ESC_IncomeDistr	Table OA.44, OA.4.2	tablesUS.escIncomeDistr
US_ESC_Voting	Table OA.45, OA.4.2	tablesUS.escVoting
US_ESC_FrenchAll	Table OA.46, OA.4.2	tablesUS.escFrenchAll
UKUS_ESC_French	Figure OA.12, OA.4.3	figuresUS.ukusEscFrench
UK_ESC_IncomeDistr	Table OA.47, OA.4.3	tablesUS.ukEscIncomeDistr
UK_ESC_Voting	Table OA.48, OA.4.3	tablesUS.ukEscVoting
UK_ESC_FrenchAll	Table OA.49, OA.4.3	tablesUS.ukEscFrenchAll
US_ESC_Path	in the paper, section 7	figuresOA.escPathPaper

Output	Where	Builder
ESC_Path	Figure OA.13, OA.4.4	figuresOA.escPath
ESC_PathTable	Table OA.50, OA.4.4	tablesOA.escPathTable
ESC_Xi	Table OA.51, OA.4.4	tablesOA.escXi
ESC_Timing	Table OA.52, OA.4.5	tablesOA.escTiming
US_ESC_ScaleWedge	Table OA.53, OA.4.6	tablesUS.escScaleWedge
NUM_Stationary	Table OA.54, OA.5.1	tablesOA.numStationary
NUM_Selection	Table OA.55, OA.5.2	tablesOA.numSelection

Built on 2026-10-07 from commit [b88d83f](#).

References

Berg, R. K., and M. Gonzalez-Eiras. 2026. “Social Security Design and Its Political Support: Technical Documentation.”